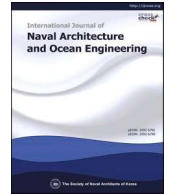




Contents lists available at ScienceDirect

International Journal of Naval Architecture and Ocean Engineering

journal homepage: www.journals.elsevier.com/international-journal-of-naval-architecture-and-ocean-engineering/

Experimental investigation on wave characteristics due to interference of catamaran demi-hulls in waves

Thi Loan Mai^a, Anh Khoa Vo^a, Aeri Cho^b, Van Thuan Mai^c, Hyeon Kyu Yoon^{d,*}^a Department of Smart Environmental Energy Engineering, Changwon National University, Changwon, Republic of Korea^b Department of Naval Architecture and Marine Engineering, Changwon National University, Changwon, Republic of Korea^c STS Engineering Co., LTD., Sejong, Republic of Korea^d Department of Smart Ocean Mobility Engineering, Changwon National University, Changwon, Republic of Korea

ARTICLE INFO

Keywords:

Wave characteristics
Model test
Catamaran
Regular waves
Irregular waves

ABSTRACT

The presence of twin hulls of a catamaran has a strong impact on wave pattern in the whole inner region due to interference between waves generated by each demi-hull. Consequently, wave profiles as well as wave height become complicated and unpredictable in the inner region not only in still water, but also in the presence of waves. Determining wave characteristics between demi-hulls of the catamaran is essential for predicting the operation of an autonomous underwater vehicle (AUV) throughout the launch and recovery (LAR) process. This study performed an experiment to investigate wave characteristics between demi-hulls of the catamaran under regular and irregular waves at various wave directions and speeds. The experiment was designed to measure wave elevation in the inner region between demi-hulls in waves using ultrasonic sensors and capacitance wave probes. The experimental method and setting were verified exactly by comparing the motion response of the catamaran measured by the Qualisys system with numerical simulation using the 3D panel method. Then, wave elevation in the inner region was measured and analyzed to wave height which is a representative parameter of regular waves and to wave spectral density and significant wave height which are representative parameters for irregular waves. These parameters were compared to those of the ocean waves to understand influences of wavelengths, sea states, wave directions, and speeds on wave characteristics between demi-hulls of the catamaran. Plotting of difference from ocean waves indicated the discrepancy in wave characteristics among positions in the inner region, wavelengths, wave directions, sea states, and speeds compared to ocean waves. Results of wave characteristics in the inner region are valuable to analyze motion response and behavior of AUV during LAR process of unmanned surface vehicle (USV).

1. Introduction

Launch and recovery (LAR) system (Fig. 1) is a process in which an autonomous underwater vehicle (AUV) and an unmanned surface vehicle (USV) collaboratively work together in a complementary mode. This system is normally done in offshore areas, remote unknown water, and denied areas where an AUV can move into ever deeper water. However, the AUV has limited endurance to travel long distances, maintain power, and communicate when it is underwater. To overcome these limitations, the USV acts as a supporting platform, playing an important role in providing the capability to greatly increase the duration and extent of AUV operation with power recharging, transport, and

data exchange among missions since it can travel long distances at high speed, execute autonomous navigations, maintain position and heading on the surface, become data and power hub to communicate with AUV, and provide the necessary need for AUV operation. The LAR process of AUV from USV is one of the most important missions in evaluating the feasibility of a collaborative autonomous system. However, the LAR system is still a huge challenge for researchers due to the disturbing environment of the open sea and complex system for fully autonomous operation which requires efficient ability, and reliable autonomous LAR of AUV and USV.

In the past, the concept design of the LAR process has been constructed, developed, and tested in multiple ways. However, researchers

Peer review under responsibility of The Society of Naval Architects of Korea.

* Corresponding author.

E-mail addresses: mailoancttt@gmail.com (T.L. Mai), anhkhoak3t@gmail.com (A.K. Vo), lee1930@naver.com (A. Cho), maivanthuan996@gmail.com (V.T. Mai), hkyoon@changwon.ac.kr (H.K. Yoon).

<https://doi.org/10.1016/j.ijnaoe.2024.100594>

Received 21 December 2023; Received in revised form 31 March 2024; Accepted 25 April 2024

Available online 3 May 2024

2092-6782/© 2024 The Society of Naval Architects of Korea. Published by Elsevier BV. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

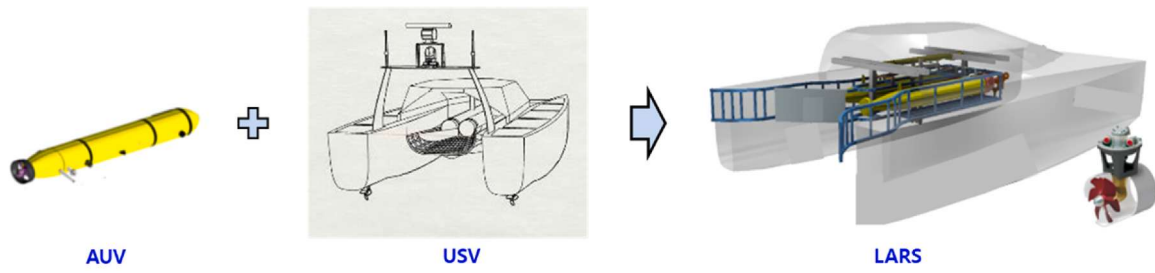


Fig. 1. AUV, USV, and LAR system.

focused on designing systems for LAR process by proposing concepts design of devices attached to USV and AUV for operating effectively (Allen et al., 2006; Pai et al., 2012; Sarda and Dhanak, 2013). The assessment of the feasibility of LAR through modeling and simulation was performed with moving USV and station-keeping USV by Sarda and Dhanak (2017, 2019), respectively. These studies ignored the impact of the environmental conditions and the inner region of the catamaran.

It is recognized that LAR mainly takes place in an open sea. It is subjected to environmental influences such as wind, waves, and currents. Moreover, a LAR is accomplished completely under the inner region between demi-hulls of the catamaran. This region has been demonstrated to be a complex region in still water by Broglia et al. (2011, 2014) with an interference wave profile of demi-hulls that can result in noticeable crests and troughs. Definitely, wave characteristics in the inner region between demi-hulls become more unpredictable as the catamaran operates in disturbing environmental conditions of the open sea such as waves, wind, and current, making AUV motion become more difficult in this area. There were studies relative to the catamaran in the past. Broglia et al. (2011, 2014), Mandru and Pacuraru (2020) and Liu et al. (2021) calculated the resistance performance with the total resistance, sinkage, and trim of the catamaran by experimental and CFD simulation methods, the interference effects were analyzed simultaneously, especially, Broglia et al. (2014) assessed it with different hull separations. Milanov et al. (2011, 2012) conducted captive model tests of the high-speed catamaran in deep and shallow waters for maneuvering and course stability purposes. The free-running model test by numerical method was also simulated to validate with system-based simulation. Pandey and Hasegawa (2016) carried out the turning maneuver of catamaran with a combined experimental and simulation methods. For catamaran in waves, Castiglione et al. (2011) predicted the seakeeping characteristics of a high-speed catamaran advancing in regular head waves using URANS method. The motion responses were determined for various speed conditions. Bouscasse et al. (2013) experimented to investigate the seakeeping behaviors with motion responses and added resistance of a fast catamaran in both head regular and irregular waves at different speed conditions. However, the previous studies of the catamaran focused on their behaviors and characteristics advancing in still water and waves.

The wave characteristics in the inner region of the catamaran travel in wave conditions are still a huge challenge. Therefore, it is necessary to investigate wave characteristics between demi-hulls of the catamaran to accurately design the LAR process for AUV. This can be achieved by measuring wave elevation in the inner region. Various types of wave height gauges have been employed for this purpose, including resistance-type wave probes (Jeon et al., 2021; Nam et al., 2022), capacitance wave probes (Nam et al., 2022; Kim et al., 2023), ultrasonic sensors (Gao et al., 2021; Jung and Koo, 2021; Kim et al., 2019, 2022, 2023), depth camera (Kim et al., 2023), and pressure transducer (Cavaleri, 1980; Bishop and Donelan, 1987). Among these, the capacitance wave probe stands out as the most common choice due to its reliability, ease of installation, and operation simplicity. Nevertheless, the capacitance wave probe utilizes an invasive approach, measuring the immersion depth of a vertical wire affixed to the support structure. Thus, it can disturb the water surface. To measure the wave elevation in the inner

region of the catamaran, an ultrasonic sensor offers an alternative method because it is a non-contact method that operates based on ultrasonic transmission and reflection principle, allowing it to accurately measure the distance between the sensor and the water surface without disturbing the water surface as confirmed by Gao et al. (2021). Therefore, the present study employed ultrasonic sensors to measure wave elevation between demi-hulls of the catamaran, while capacitance wave probes are used to measure wave elevation of ocean waves.

In this study, an experiment was designed and conducted in the presence of a catamaran to measure wave elevation in the inner region between demi-hulls of the catamaran. The experiment covered both regular and irregular waves to investigate wave characteristics under various wave conditions and different speeds of catamaran. The experimental method and setting were verified by comparing the motion response of the catamaran in regular waves with the numerical calculation. For wave elevation, The signal in regular waves was described as harmonic motion, enabling wave height to be determined using the Fourier series as presented by Mai et al. (2023). On the other hand, for irregular waves, the wave spectral density was analyzed using fast Fourier Transform (FFT) method to decompose the complex wave elevation signal into its constituent frequencies and amplitudes as done by Aggarwal et al. (2018) and Oh et al. (2020). Subsequently, significant wave height was calculated based on wave spectral density. Wave characteristics in the inner region between demi-hulls of the catamaran were analyzed and discussed by comparing them to ocean waves. This comparison provides insights into how wave behavior differs in the inner region under different wave conditions. Results of wave characteristics in the inner region hold significant value in assessing motion response and behavior of AUV during LAR process to USV.

2. Experimental procedures

2.1. Model test

A catamaran, known as the Kumha catamaran, was constructed with two identical demi-hulls interconnected through robust support structures used for LAR purpose. The model was scaled with 1/7.94 scale

Table 1
Main specifications of catamaran.

Description	Unit	Full-scale	Model-scale
Scale ratio, λ		1	1/7.94
Length overall, L_{OA}	m	12.00	1.511
Length between perpendiculars, L_{pp}	m	11.91	1.500
Beam overall, B	m	6.00	0.756
Beam demi-hull, b	m	1.50	0.189
Draft, T	m	0.65	0.082
Distance between center of hulls, H	m	4.50	0.563
Displacement weight, W	kgf	16,062	31.775
Displacement volume, Δ	m ³	15.67	0.031
LCG from FP	m	5.60	0.705
VCG	m	1.10	0.139
Roll radius of gyration, k_{xx}	B	0.28	0.28
Pitch radius of gyration, k_{yy}	L_{pp}	0.25	0.25
Yaw radius of gyration, k_{zz}	L_{pp}	0.25	0.25

ratio into 1.5 m in length between perpendiculars, enabling testing in the limited tank. Table 1 summarizes main specifications for both full-scale and model-scale of the catamaran. Fig. 2 visually presents the catamaran model that built for this study.

2.2. Experimental facility and instrumentation

Wave height was measured in a 3-dimensional wave basin situated at CWNU (Changwon National University), which was 20 m in length, 14 m in width, and 1.8 m in depth as shown in Fig. 3. This facility consisted of a range of essential equipment such as a wave generator, wave absorber, and towing carriage. The wave generator was the piston type assembled with an array of 28 pistons, which was installed at the end of

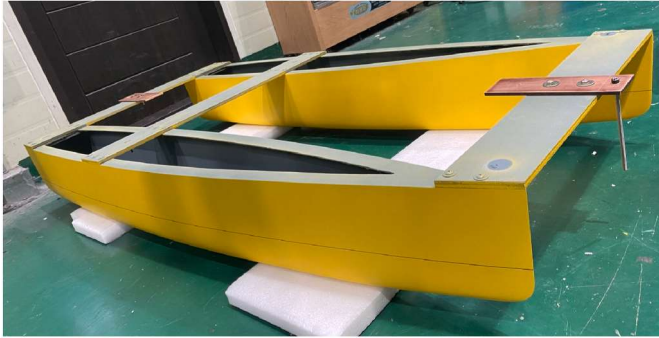


Fig. 2. Catamaran model.

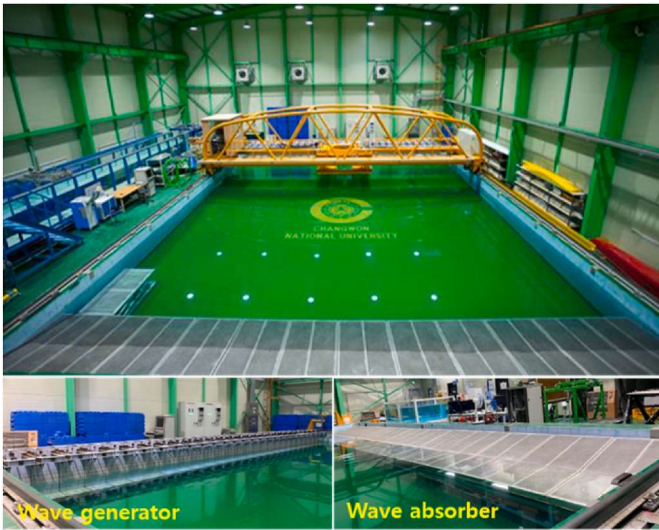


Fig. 3. Experimental facility.

the tank. This wave generator could generate both regular and irregular waves based on standards of the International Towing Tank Conference (ITTC), Neumann, Pierson-Moskowitz (PM), Jount Nort Sea Wave Project (JONSWAP), and transformation (TMA). To minimize undesirable wave reflections and reduce waiting time for each test, a wave absorber was positioned on opposite sides of the wave generator. To implement movement of a ship model, a towing carriage with sub-carriage and yaw table was placed in the tank. Towing carriage enabled towing the ship forward and backward in the x-direction using two rails at the sides of the tank. Sub-carriage could move to the right and left sides in the y-direction by two rails on the towing carriage. Yaw table was a part of the sub-carriage which could rotate to the right and left sides. Combining functionalities of the towing carriage, sub-carriage, and yaw table allowed for carrying out the test at forward speed and different wave directions. Given that the wave generator made the wave in a single direction, the test for different wave directions was conducted by changing towing directions in forward, diagonal, transverse, and backward directions as shown in Fig. 4.

Wave height and free surface elevation over time were monitored using wave height gauges equipped with ultrasonic sensors and capacitance wave probes. The ultrasonic sensor uses a non-contact method based on the principle of emitting and receiving high-frequency sound waves to measure distance as described visually in Fig. 5. The sensor emits a short burst of ultrasonic waves in the form of sound waves that travel through the air. Upon encountering an object, waves bounce back. The sensor's transducer then switches to receive reflected waves. The capacitance wave probe uses a contact method based on the principle of changes in capacitance due to the presence of fluid surface. The probe is represented as a thin insulated conducting wire. When the probe is partially submerged in water, the wave height is measured by detecting changes in capacitance of the wire as waves propagate and pass through the probe as shown in Fig. 5. The wave height signal of the ultrasonic sensor and capacitance wave

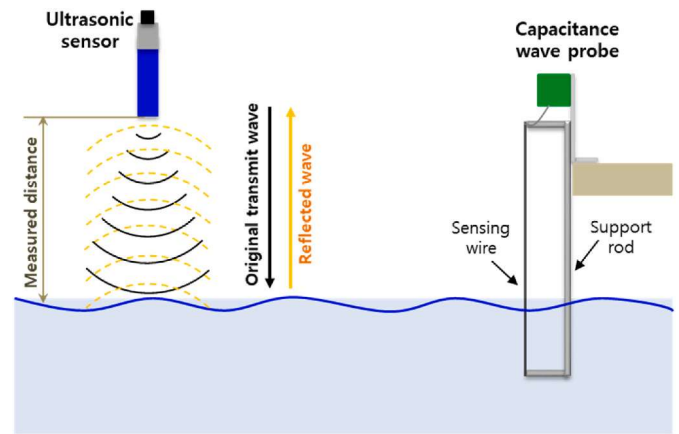


Fig. 5. Principles measurement of wave height gauges.

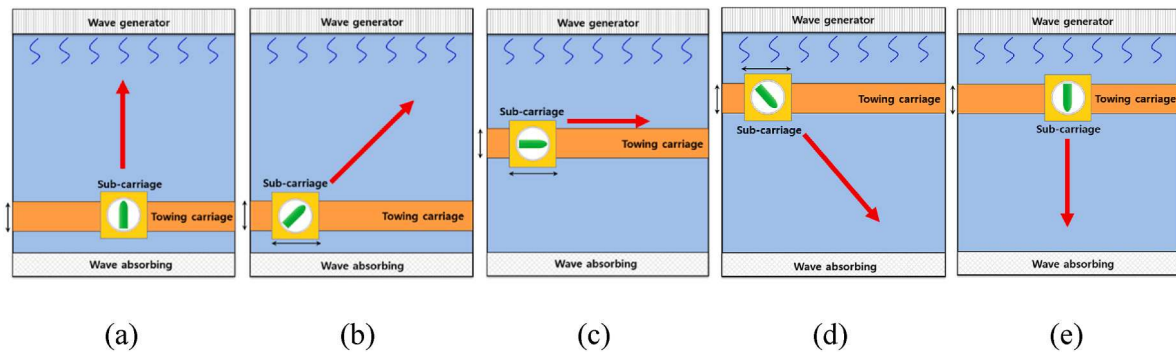


Fig. 4. Movement of ship for various wave directions: (a) Head waves; (b) Bow quartering waves; (c) Beam waves; (d) Stern quartering waves; (e) Following waves.

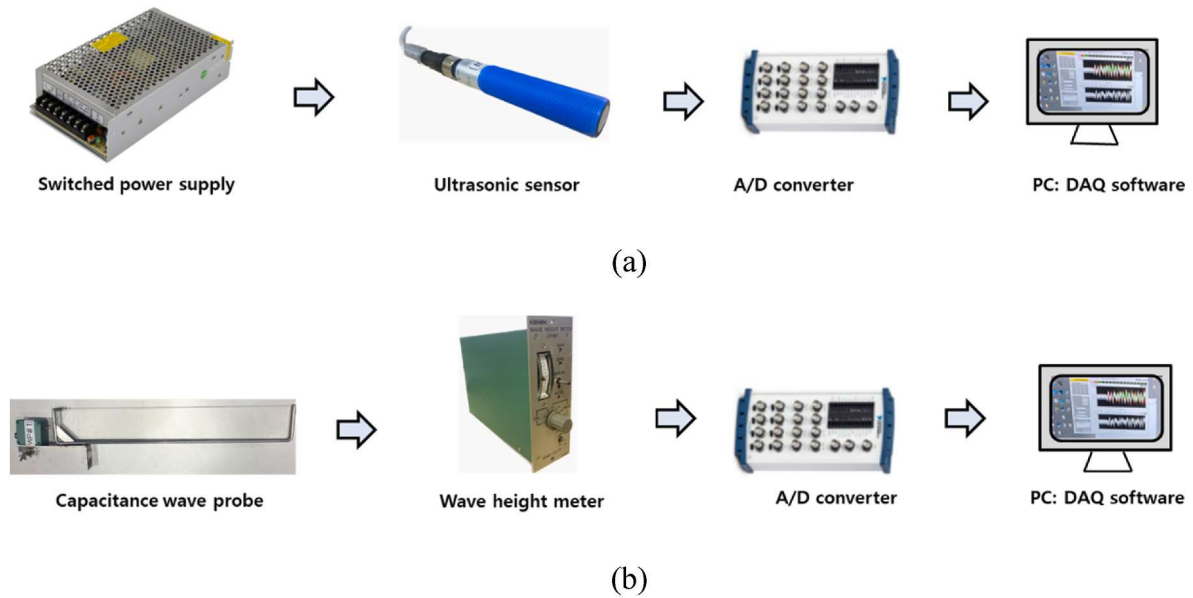


Fig. 6. Data acquisition process for wave height gauges: (a) Ultrasonic sensor; (b) Capacitance wave probe.

probe is taken simultaneously using data acquisition (DAQ) software, although their systems operate independently. For data acquisition of the ultrasonic sensor, besides main power, a switched power supply 12 V is employed to convert the electric current that comes from a power source to the voltage value which provides electrical power to the ultrasonic sensor. The response wave height from the ultrasonic sensor is transferred directly to the analog-to-digital (A/D) converter as exhibited in Fig. 6a. In a different way from the ultrasonic sensor, the measurement of the wave height from the capacitance wave probe (WP) is transmitted to the wave height meter. It then moves to the A/D converter as presented in Fig. 6b. Both wave height gauges can transmit through the A/D converter. Their digital signals are recorded on a computer with DAQ software as displayed visually in Fig. 6. On the other hand, although the capacitance wave probe

gives results with high precision, it is immersed in water which interferes with the ship's movement and disturbs water surface due to wave patterns generated by the support rob, even damaging the wave probe when measuring wave height at restricted positions between demi-hulls of the catamaran. Therefore, the non-contact method of ultrasonic sensor is the best choice to measure wave height between demi-hulls.

2.3. Pre-test of model ship and instrument

When performing test with the model ship in waves, it is crucial to make sure that the model ship's features are similar to those of a real ship. To achieve this, a series of pre-tests were executed on the model ship prior to experimenting with it in waves. Fig. 7 shows a pre-test of

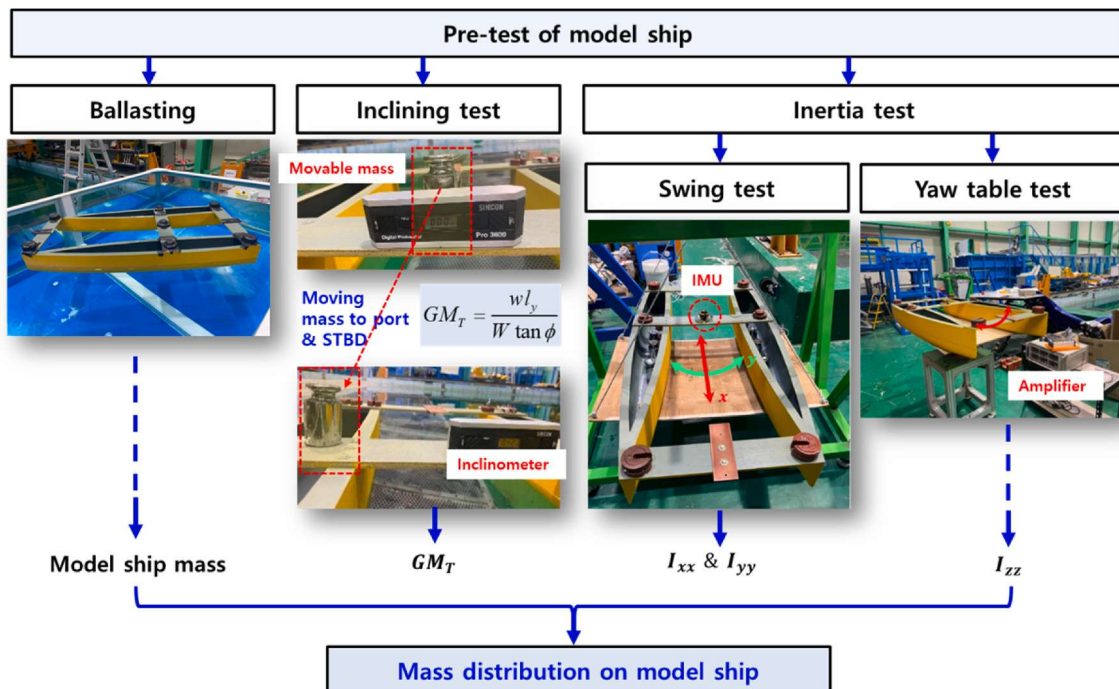


Fig. 7. Pre-test of model ship.

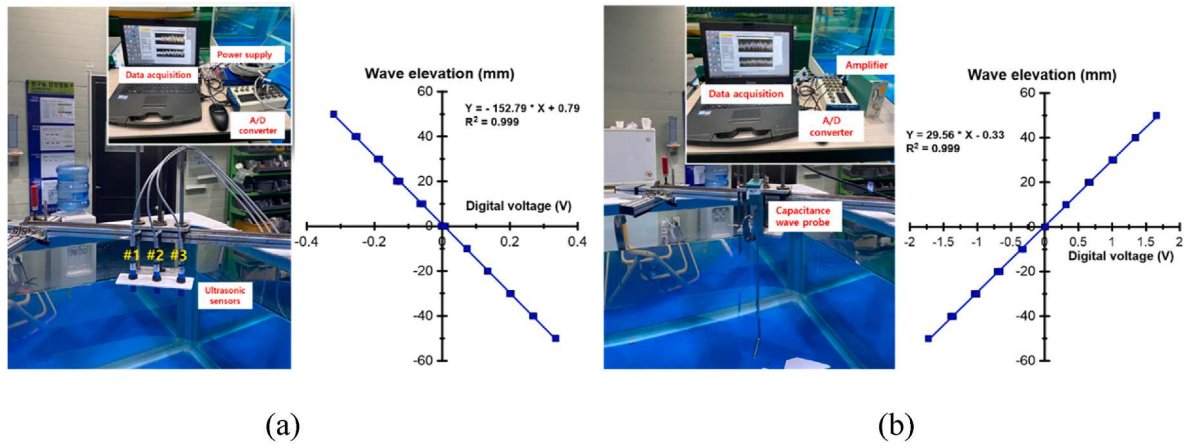


Fig. 8. Calibration of wave height gauges: (a) Ultrasonic sensor; (b) Capacitance wave probe.

Table 2

Wave conditions for regular waves.

λ/L_{pp} (—)	Wavelength, λ (m)	Target wave height, H_w (mm)	Wave frequency, ω (rad/s)	Wave period, (s)	Input wave height, (mm)
0.50	0.750	15.0	9.066	0.69	16.2
0.75	1.125	22.5	7.402	0.85	24.0
1.00	1.500	30.0	6.410	0.98	31.0
1.25	1.875	37.5	5.734	1.10	40.0
1.50	2.250	45.0	5.234	1.20	49.0
1.75	2.625	52.5	4.846	1.30	59.0
2.00	3.000	60.0	4.533	1.39	70.0

the model ship, which comprises ballasting, inclining, and inertia tests. These tests serve to confirm the model ship mass, transversal meta-centric height (GM_T), and radius of gyrations, respectively. For ballasting, the sum of supplementary weight and bare hull's weight was 31.775 kgf. The draft should simultaneously reach the black line on the model located 0.082 m from the baseline. The inclining test was used to check whether the mass distribution was proper with GM_T . The movable mass was moved to the starboard and port to measure the heel angle using an inclinometer. Similarly, the inertia test was conducted to check the radius of gyrations of roll, pitch, and yaw. Mass moment of inertia was a quantity representative for the radius of gyration calculated by period which was measured using a swing table and a yaw table. Resulting

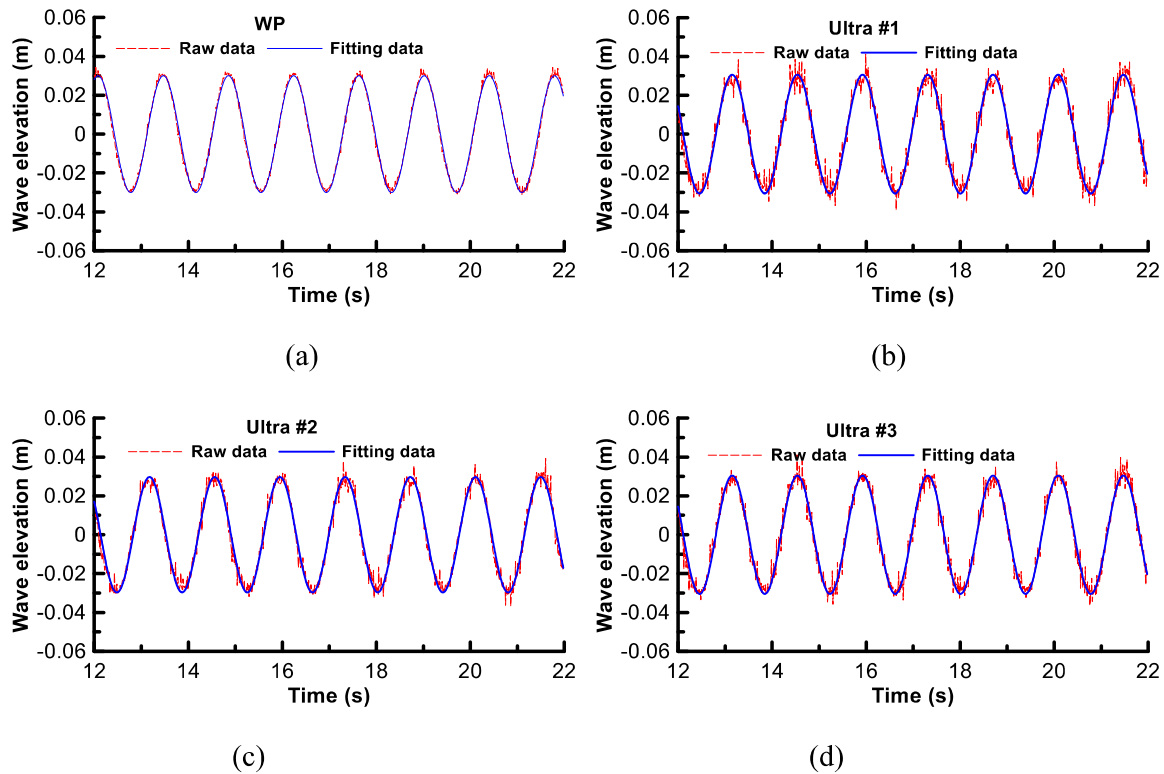


Fig. 9. Fitting curve of wave signal for wave measurement instruments at $\lambda/L_{pp} = 2.0$: (a) Capacitance wave probe; (b) Ultrasonic #1; (c) Ultrasonic #2; (d) Ultrasonic #3.

values had an error within 5% compared to theoretical calculation. All pre-tests were implemented to confirm that the mass distribution on the model ship accurately reflected the features of the real ship.

Experimental efficiency primarily depends on precision of measurements for obtaining reliable and meaningful data. Therefore, calibrating measurement instruments is needed. Fig. 8 shows calibrations of the ultrasonic sensor and capacitance wave probe in still water. Wave height gauges were attached to a frame that could adjust vertically to get an output digital signal corresponding to the distance from the ultrasonic to the water surface of the ultrasonic sensor. The wave probe position was submerged in water of the capacitance wave probe. Consequently, calibration factors of ultrasonic sensors and capacitance wave probes were taken to convert the output digital signal into wave height measurements.

2.4. Calibration of wave generator

Wave height generated by a wave generator in the wave basin is also an important parameter. Waveform as well as wave height are formed from the wave generator through a software interface by inputting wave height and frequency. However, wave height depends on water depths in the tank. Additionally, the distance between the wave generator and wave height gauges was 6.4 m with wave height gauges normally mounted on the carriage located at the center of the tank during calibration of the wave generator. Calibration of wave height generated from the wave generator was conducted by adjusting wave height to find out the input wave height corresponding to the target wave height for each frequency. Target wave height and frequency were determined based on the ratio of wavelength (λ) to length between perpendiculars (L_{pp}) ranging from 0.5 to 2.0. The ratio of wave height to wavelength was 1/50 recommended by ITTC (2021) for regular wave conditions.

Table 2 lists conditions for regular waves. λ/L_{pp} ranged from 0.5 to 2.0 with target wave height and frequency correspondence. The wave height signal of each measurement instrument in regular waves at λ/L_{pp} of 2.0 as an example is presented in Fig. 9. Because the wave elevation was developed to be greater gradually since marking a wave until a constant, so, it had to subtract the initial periods and used the data at the constant of wave elevation by taking the time series from 12 s to 22 s as shown in Fig. 9. It can be seen that the raw signal of the capacitance wave probe is smoother than that of the ultrasonic sensors because the capacitance change is directly related to the distance, whereas reflected sound waves are influenced by various factors of the environment. However, the wave height value of the capacitance wave probe and that of the ultrasonic sensor were observed to be similar (Fig. 9). To obtain wave height value of regular waves, the wave signal was described as a Fourier series in Eq. (1), where a_0 , a_1 , and a_2 were Fourier coefficients presented as a vector $\underline{\beta}$ that could be taken by calculating Eq. (2), where H was an independent variable matrix in Eq. (3) and vector $\underline{Z}$ in Eq. (4) was the measured wave signal. Following the procedure, the fitting curve of wave signals was taken as displayed in Fig. 9 for wave measurement instruments at $\lambda/L_{pp} = 2.0$. It could be seen that fitting curves were filtered quite smoothly compared to raw data, especially for ultrasonic sensors. Wave height values were then determined for measurement instruments.

Table 3

Wave conditions for irregular waves of scale model.

Sea state	Target significant wave height, $H_{1/3}$ (mm)	Modal wave period, T_p (s)	Input significant wave height (mm)
SS2	37.8	1.73	37.5
SS3	110.8	2.06	108.9

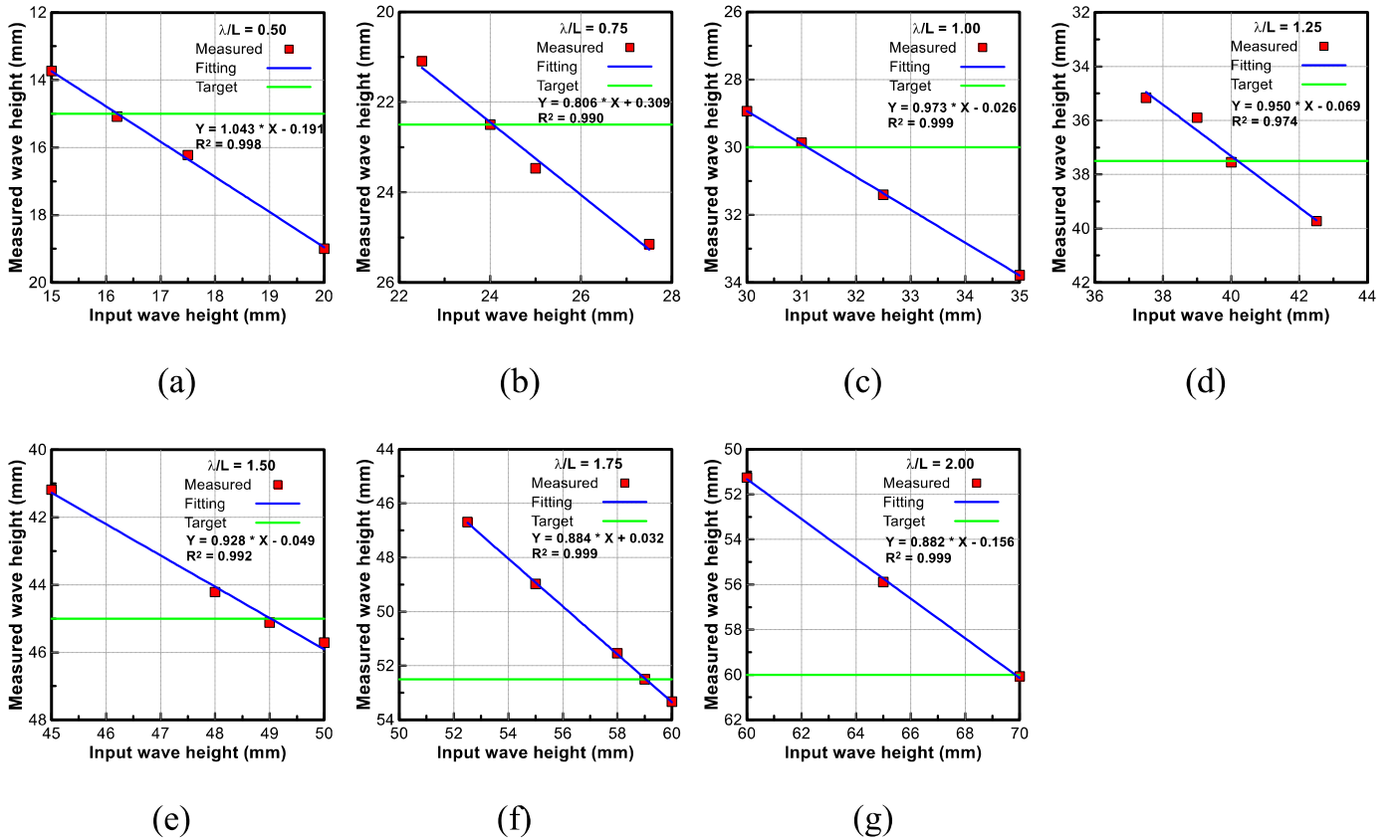


Fig. 10. Calibration of wave generator in regular waves: (a) $\lambda/L = 0.5$; (b) $\lambda/L = 0.75$; (c) $\lambda/L = 1.0$; (d) $\lambda/L = 1.25$; (e) $\lambda/L = 1.5$; (f) $\lambda/L = 1.75$; (g) $\lambda/L = 2.0$.

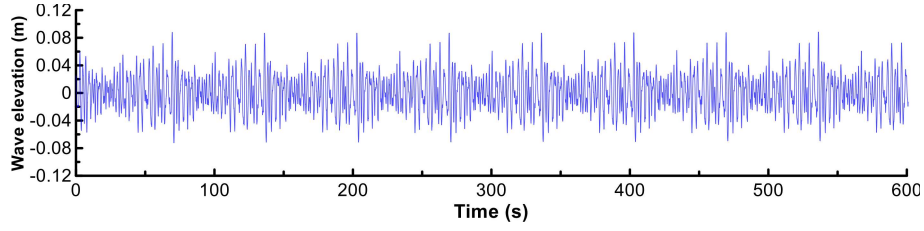


Fig. 11. Measured wave elevation at SS3 within 10 min.

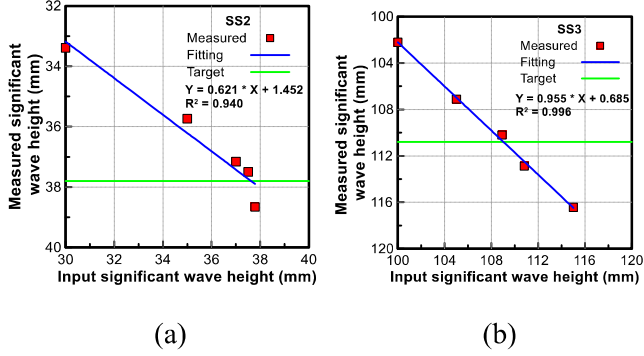


Fig. 12. Calibration of wave generator in irregular waves: (a) SS2; (b) SS3.

$$z(t) = a_0 + a_1 \cos(\omega t) + a_2 \sin(\omega t) \quad (1)$$

$$\underline{\beta} = (H^T H)^{-1} H^T \underline{Z} \quad (2)$$

$$H = \begin{bmatrix} 1 & \cos \omega t_1 & \sin \omega t_1 \\ 1 & \cos \omega t_2 & \sin \omega t_2 \\ \vdots & \vdots & \vdots \\ 1 & \cos \omega t_n & \sin \omega t_n \end{bmatrix} \quad (3)$$

$$\underline{Z} = [z(t_1) \quad z(t_2) \quad \dots \quad z(t_n)] \quad (4)$$

Uncertainty analysis of wave height in regular waves with constant wave period was conducted for each λ/L_{pp} to figure out the exact input wave height entered from the wave generator to be identical to the target wave height measured by wave measurement instruments. Fig. 10 shows results of calibration of the wave generator in regular waves for different λ/L_{pp} . Based on a linear regression model, input wave heights entered from the wave generator were determined. Results are listed in Table 2.

In a similar manner, calibration of the wave generator for irregular waves is executed with different significant wave heights ($H_{1/3}$) and constant modal wave period (T_p) of each sea state. Lewis (1989) has provided information for significant wave height and model wave period corresponding to each sea state. Table 3 depicts wave conditions for irregular waves with $H_{1/3}$ and T_p of SS2 and SS3. Fig. 11 shows an example of measured wave elevation at SS3 within 10 min.

The irregular wave signal was generated by the superposition of multiple regular wave components, each with its own amplitude, frequency, and period of propagation. Consequently, the surface wave elevation could be expressed as a sinusoidal form in Eq. (5),

$$\zeta(x, t) = \sum_{n=1}^N a_n \cos(\omega_n t - k_n x + \varepsilon_n) \quad (5)$$

where, a_n , ω_n , k_n , and ε_n are the amplitude, frequency, wave number, and phase of the n th component in regular waves, respectively. N is the number of components.

Applying Fourier transform for the irregular wave signal, the surface wave elevation can be rewritten as Eq. (6):

$$\zeta(t) = \sum_{n=1}^N (a_1^n \cos \omega_n t + b_1^n \sin \omega_n t) \quad (6)$$

Using fast Fourier transform (FFT) to decompose the complex wave signal from the time domain into its constituent frequencies and amplitudes, the wave energy spectrum ($S_\zeta(\omega)$), can be expressed as in Eq. (7):

$$S_\zeta(\omega_n) \cdot d\omega = \frac{1}{2} \zeta_{a_n}^2 \quad (7)$$

where m implies a moment of properties of the wave spectrum with respect to the vertical axis at $\omega = 0$ and $m_{n\zeta}$ indicates the n th-order moment written as Eq. (8):

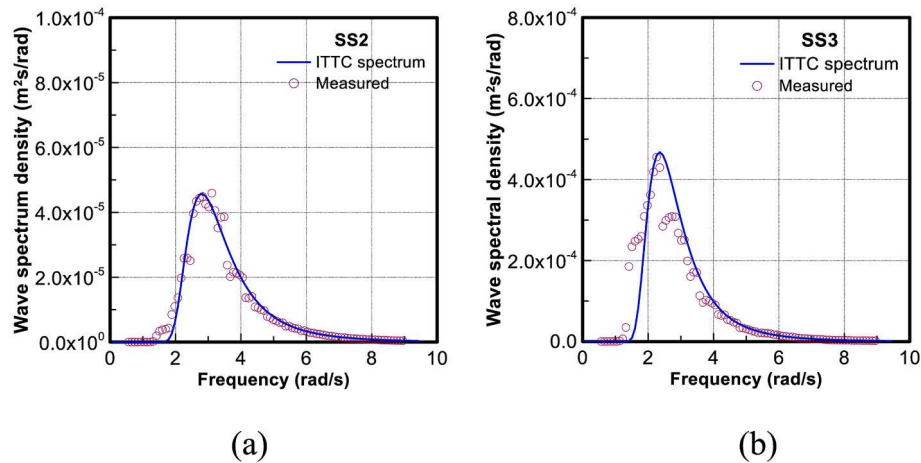


Fig. 13. Wave spectral density confirmation by wave generator: (a) SS2; (b) SS3.

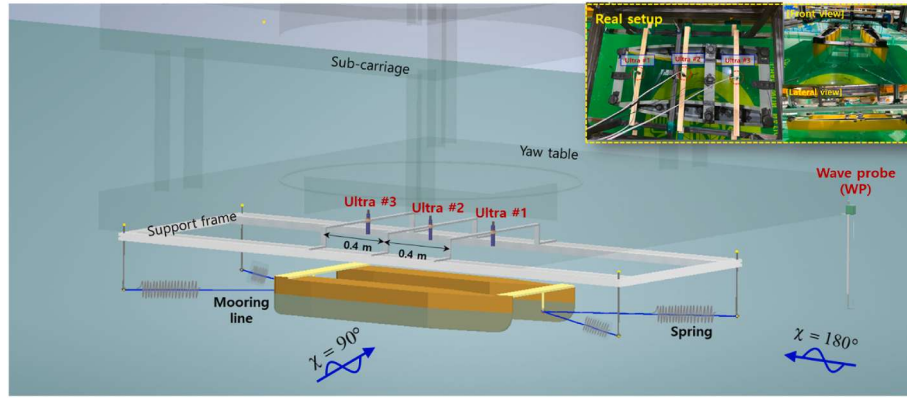


Fig. 14. Schematic of experiment.

$$m_{n\zeta} = \int_0^{\infty} \omega^n \cdot S_{\zeta}(\omega) \cdot d\omega \quad (8)$$

Significant wave height is then calculated with Eq. (9):

$$H_{1/3} = 4\sqrt{m_{0\zeta}} \quad (9)$$

where $m_{0\zeta}$ means the area under the spectral curve. It is an indication of the variance squared (σ_{ζ}^2) as presented in Eq. (10):

$$m_{0\zeta} = \sigma_{\zeta}^2 = \int_0^{\infty} S_{\zeta}(\omega) \cdot d\omega \quad (10)$$

Significant wave height was calculated in Eq. (9). It was used as a target parameter to calibrate the wave generator. Fig. 12 shows

calibration results of the wave generator in irregular waves by varying the significant wave height for each sea state. Based on the estimated linear regression model, input significant wave heights entered from the wave generator are found as presented in Table 3. Input significant wave heights were reaffirmed by comparing the wave spectral density generated by the wave generator with those calculated by the ITTC formula as described in Eq. (11). Fig. 13 displays wave spectral density confirmed by the wave generator at SS2 and SS3. It showed fairly good agreement with the wave density calculated by the ITTC formula. Therefore, input significant wave heights were confirmed to be proper:

$$S(\omega) = \frac{A}{\omega^5} \exp\left(-\frac{B}{\omega^4}\right) \quad (11)$$

here, $A = 173 \frac{H_{1/3}^2}{T_1^4}$, $B = 691 \frac{1}{T_1^4}$

2.5. Experimental setup

Experiment measuring wave height between demi-hulls of the catamaran was performed under both regular and irregular waves. After ensuring features of the model test in pre-test and completing calibration of measurement instruments, model test and measurement instrument were installed in the wave basin. Fig. 14 shows a schematic of the experiment in a 3-dimensional wave basin with the catamaran model and arrangement of wave measurement instruments. The catamaran model was connected to a towing carriage by a mooring line system. A support frame enabled the catamaran model to move in multiple directions. The mooring line system consisted of unstretched cable and spring linked symmetrically at four corners, enabling the catamaran model to flexibly respond six degrees of freedom (DoF) motions. On the other hand, a capacitance wave probe (WP) was located outside at the sub-carriage to measure wave height representative of the ocean waves. Meanwhile, three ultrasonic sensors (Ultra) were fixed on small frames distance of 0.4 m positioned at the middle region of demi-hulls to

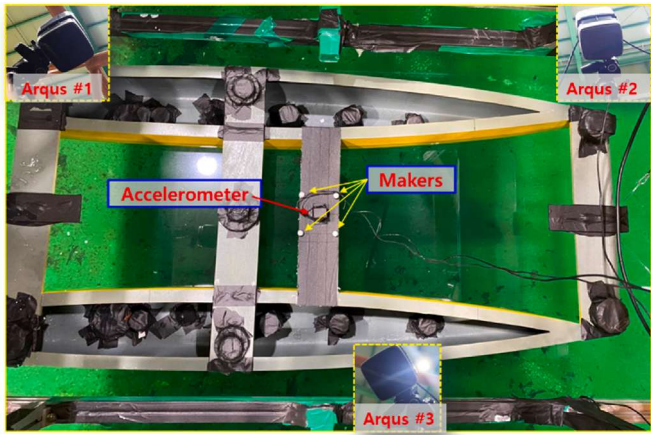


Fig. 15. Experimental setup and measurement sensors for ship motion.

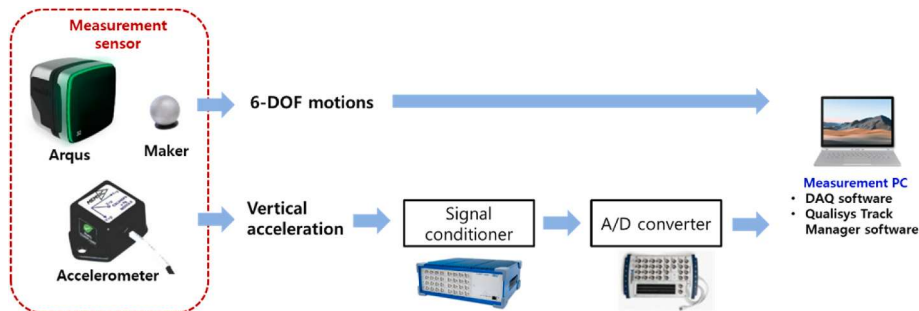


Fig. 16. Data acquisition process for ship motion measurement.

measure the wave height in the inner regions of catamaran. The 0.4 m spacing among the ultrasonic sensor was decided based on the length of the model ship of 1.5 and strong wave interference with trough and peak waves occurred at distance by a quarter along center of hull (Broglia et al., 2011, 2014). The ultrasonic sensor positions were decided to be far enough from each other and far from other electrical devices to minimize the interference between neighbor devices. Moreover, the ultrasonic sensors were placed at the appropriate height within the allowable sensing range from 0.08 to 1.6 m recommended by the manufacturer to record the signal well. On the other hand, it was recognized that it would be more reasonable to measure the wave elevation in the inner region at a certain position relative to the model, which meant that the ultrasonic installed in the model could travel with it. Nevertheless, limitations in the size of the wave basin and the model ship made it impossible to install ultrasonic sensors on the model ship, leading to errors in wave signal due to the model ship's strong movement and insufficient sensing distance of ultrasonic sensors. For this reason, we mounted the ultrasonic sensor on the carriage to measure the wave elevation at a specific position relative to the carriage, which still fell in the inner region between the demi-hulls of catamaran during its movement.

3. Results and discussion

3.1. Validation of experimental method

Besides the wave height measurement, the motion response of the

ship in regular waves was measured to verify the experimental method and setting. The motion responses were measured by Qualisys system which used high-speed cameras named “Arqus” and markers to track the movement of the ship. The markers were attached to the mid-ship and it was typically the retroreflective sphere that reflected light back to the camera, marking it easily detectable in the camera images. Moreover, the vertical acceleration was measured by an accelerometer located at mid-ship to analyze and interpret the phenomenons of wave elevation and ship motion. Fig. 15 shows the experimental setup and measurement sensors of the accelerometer and markers for the ship motion. The signal of the 6-DoF motions was directly output on PC by Qualisys track manager software, while the signal of acceleration needed a communication bridge between sensors and computer which are signal conditioner and A/D converter (see Fig. 16).

Figs. 17–19 show comparison results of motion RAOs (Response Amplitude Operator) measured by experiment and numerical calculation for typical wave directions at zero-speed condition. The numerical method is calculated based on the 3D panel method in the potential code of Ansys AQWA. It can be seen that the experiment and numerical calculation for heave RAO, roll RAO, and pitch RAO at various wave directions agree relatively well. This demonstrated the precision of the experimental method and setting. In addition, the motion RAOs at forward speed are simultaneously plotted in Figs. 17–19 to inspect the difference in motion response between forward speed and zero speed. Because of low-speed condition, the motion RAO is quite similar to those of zero-speed, especially at long wavelengths. Moreover, the vertical accelerations at zero-speed and forward-speed conditions are measured

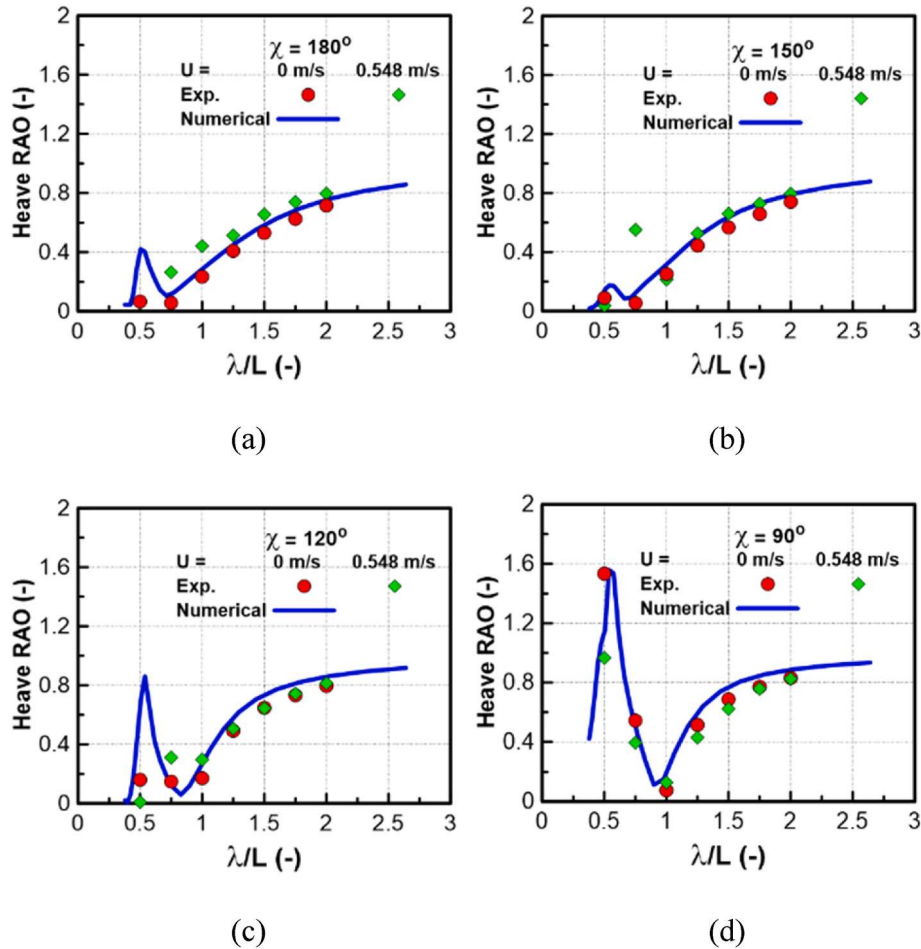


Fig. 17. Heave RAO: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$.

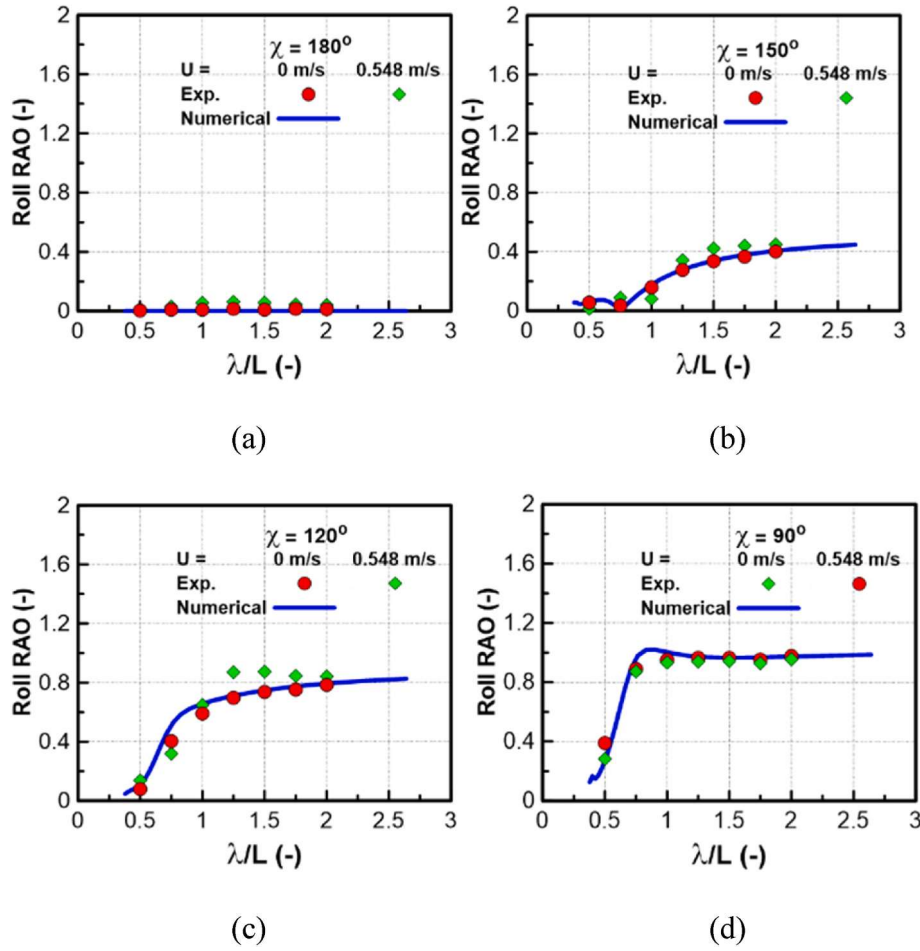


Fig. 18. Roll RAO: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$.

and depicted in Fig. 20 for typical wave directions. The vertical acceleration responses of forward-speed are seen to be higher than that of zero-speed at head waves and bow quartering waves, and become similar at beam waves. The RAO motion and vertical acceleration responses are used to interpret and analyze the wave height phenomena in the next section.

3.2. Regular waves

In the case of regular waves, wave characteristics in the inner region between demi-hulls of the catamaran were investigated at different positions, wavelengths, wave directions, and speeds. Figs. 21 and 22 illustrate time histories of measured wave elevation in regular waves at zero speed and forward speed of 3 knots for full scale corresponding to 0.548 m/s for model-scale. Observing Fig. 21 for wave elevation at zero speed, the wave elevation in the inner region at long wavelength ($\lambda/L = 1.75$) appeared to exhibit insignificant changes for all wave directions in general. Upon closer examination, a detailed review revealed that the wave elevation did experience a slight change compared to ocean waves of the capacitance wave probe due to wave interference between demi-hulls of the catamaran. For instance, surface waves in the inner region were more concave. This phenomenon became increasingly pronounced as the wave direction approached 90° . Simultaneously, it also rose

slightly higher than ocean waves, resulting in an overall increase in wave elevation as the wave direction advanced to 90° . The increase in wave elevation at the wave directions of quartering waves ($\chi = 150^\circ$, 120° , 60° , 30°) and beam waves ($\chi = 90^\circ$) was attributed to the supplementation of radiation waves generated by the hull, were added to the surface wave elevation due to wave interaction between demi-hulls and the ship's movement. For shorter wavelength ($\lambda/L = 1.0$), the wave elevation has a huge change depending on the measurement position in the inner region. Due to a smaller wave height of a shorter wavelength, wave elevations in the inner region at wave directions of head wave ($\chi = 180^\circ$) and following wave ($\chi = 0^\circ$) were smaller than those of ocean waves. It ascended as the wave direction approached 90° . In particular, for surface waves at ultrasonic 2 and 3 positions of bow waves and ultrasonic 1 and 2 positions of stern waves, the wave elevation experienced a considerable increase for both concave down and convex up. Meanwhile, the surface wave of ultrasonic 1 position exhibited a convex up, but only at beam waves. Therefore, wave elevation in the inner region not only depends on wave interference between demi-hulls, but also depends on wave direction which is the reason for radiation waves generated by the hull.

Unlike zero speed, wave elevation in the inner region with catamaran moving at a forward speed of 0.548 m/s had significant changes for both long and shorter wavelengths as illustrated in Fig. 22. The

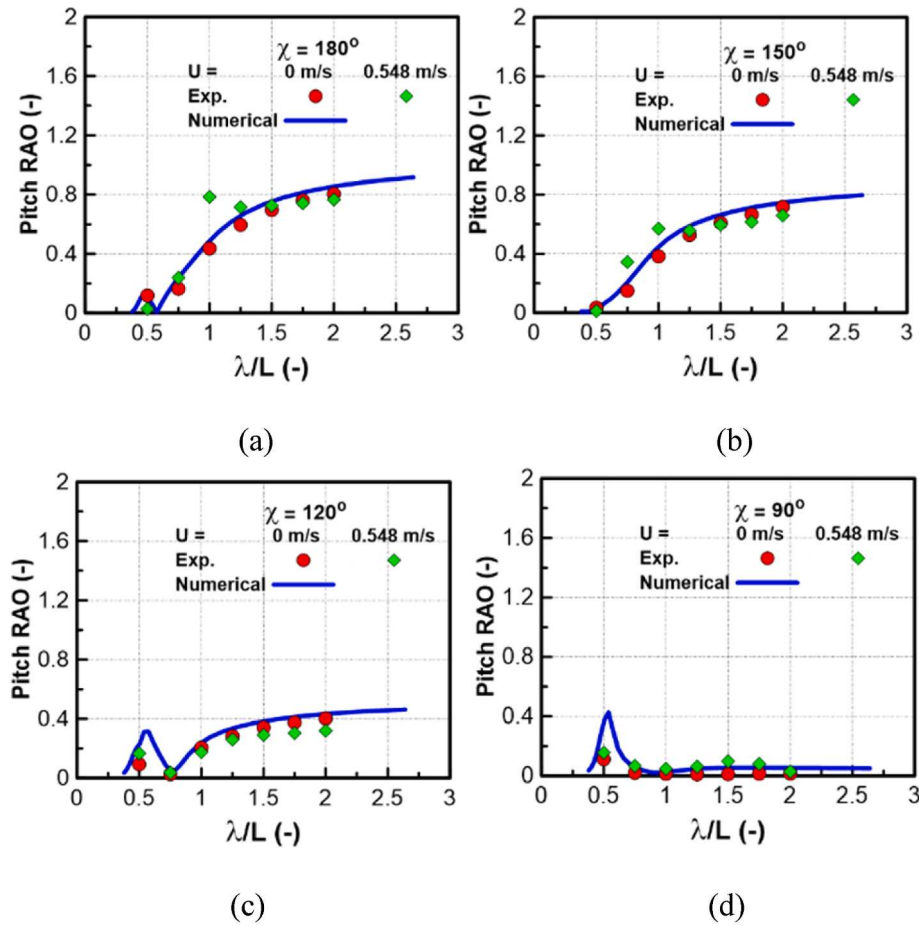


Fig. 19. Pitch RAO: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$.

higher vertical acceleration response of ship motion at forward speed as shown in Fig. 20 was one of the factors that contributed to changes in wave elevation. This led to generating the strong wave radiation from the ship hull which cohered to wave interference in the inner region to generate more concave down and convex up of waves. For long wavelength ($\lambda/L = 1.75$), it could be seen that wave elevation at ultrasonic 1 position took on a deeply concave down in head wave ($\chi = 180^\circ$) and bow quartering wave ($\chi = 150^\circ$), whereas, wave elevation at ultrasonic 2 and 3 positions was smaller than that of ocean wave in head waves and following waves. It ascended as wave direction approached 90° . Regarding shorter wavelength ($\lambda/L = 1.0$), wave elevations at ultrasonic 1, 2, and 3 positions were observed to be the largest at wave directions of 30° , 180° , and 60° , respectively. On the other hand, they were found to be the smallest at a wave direction of 0° . Consequently, wave height at the inner region between demi-hulls of catamaran did not have straightforward rules to predict due to the complex interaction between waves and ship's geometry, especially at forward speed and shorter wavelength. Additionally, Fig. 23 visualizes wave interference in the inner region at forward speed and wave direction of 180° during regular waves. It could be seen that the divergent wave interval of $\lambda/L = 2.0$ was greater than that of $\lambda/L = 1.5$. Wave interference of $\lambda/L = 2.0$ was also stronger than that of $\lambda/L = 1.5$, which emphasized the effect of forward speed on wave elevation in the inner region. Therefore, wave elevation in this region was influenced by various factors including wavelength, wave direction, and speed.

Through analysis of time series of measured wave elevation, wave heights in the inner region between demi-hulls of regular waves were

calculated following the procedure outlined in Eqs. (2)–(4). Figs. 24 and 25 comprehensively summarize wave height in the inner region of the catamaran considering various positions, wavelengths, wave directions, and speeds. To visualize the effects of wave interference, wavelengths, wave direction, and speed on wave height in inner region, the difference in wave height compared to ocean waves was also graphed. A negative difference value indicated that the wave height value in the inner region was smaller than that of ocean waves, while a positive difference value indicated that the wave height value was higher than that of ocean waves. For zero speed of the catamaran in Fig. 23, the wave height in the inner region closely resembled that of ocean wave for long wavelengths with relatively small differences at the head wave and following wave. It increased to be higher than ocean waves at wave direction towards 90° with difference values smaller than 20%. In shorter wavelengths, the wave height in the inner region presented higher unpredictability. It significantly differed from the wave height of ocean waves at specific wave directions, such as 60° with $\lambda/L = 0.75$ and 0.5 , and at 120° with $\lambda/L = 0.5$. In these cases, the differences between the inner region and ocean waves exceed 80% while differences for other wave directions were within 40%.

Fig. 25 indicates the influence of the forward speed on wave height in the inner region between demi-hulls. Notably, there were substantial disparities in wave height at various positions in the inner region and different wavelengths. Specifically, at ultrasonic 1 position, the wave height was the greatest in head waves and the second greatest in bow quartering waves for λ/L of 2.0, 1.75, 1.5, and 0.5, with differences reaching up to 100%. However, the wave height at ultrasonic 2 position

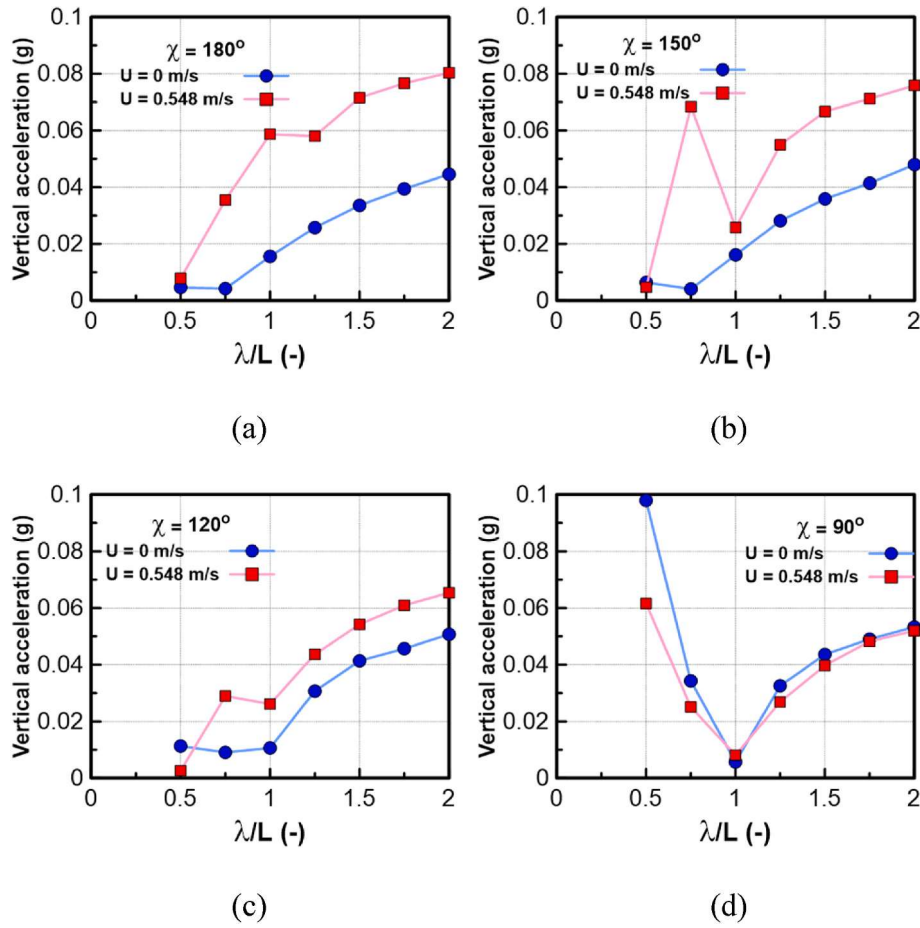


Fig. 20. Vertical acceleration: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$.

reached its maximum for λ/L of 1.0 (difference of 87%) in head waves. It reached the highest at the ultrasonic 3 position when λ/L was 0.75 with a difference of 132%. This phenomenon was a result of wave interference in combination with wave radiation generated by the ship hull at forward speed creating concave down waves in head waves. Not only head waves and bow quartering waves, but also the wave heights in other wave directions showed big changes without any discernible rule. For instance, wave height values of ultrasonics 2 and 3 reached the crest at a wave direction of 90° in long wavelengths or 60° and 30° in shorter wavelengths. Conversely, wave heights for following waves were observed to be the lowest. Furthermore, even in the same wave direction, wave height values exhibit considerable disparities among ultrasonic positions. Consequently, the combination of wave interference, wave radiation from the hull, and catamaran's forward speed contributed to the unpredictability and complexity of wave heights in the inner region between demi-hulls of catamaran.

3.3. Irregular waves

Unlike regular waves, wave characteristics of irregular waves in the inner region between demi-hulls were specifically studied at SS2 and SS3 with wave conditions outlined in Table 3. To assess effects of catamaran's presence on inner region wave characteristics in irregular waves, representative parameters such as wave spectral density and

significant wave height were analyzed. Figs. 26 and 27 display time histories of measured wave elevation of SS2 and SS3 with the first 120 s at zero speed for various wave directions. In analysis data, wave elevation was measured continuously for 10 min. It was observed that wave elevations between the inner region and ocean waves appeared to be nearly equivalent in wave height. However, due to the intense movement of the ship in irregular waves, particularly in quartering waves and beam wave directions, the ship underwent substantial translation and rotation. This led to disturbance in the inner region, causing a shift in wave frequency and wave height in the inner region compared to ocean waves. Fig. 28 visualizes wave interference in the inner region at zero speed during irregular waves of SS3. Examining Fig. 28a for a wave direction of 180° , waves moved according to the surging movement. However, alteration was insignificant with ocean waves. On the other hand, waves in the inner region of the wave direction of 90° were notably changed (Fig. 28b) when it mutually impacted between demi-hull due to a large movement of the ship in sway and yaw. This demonstrated that the wave elevation in the inner region during irregular waves depended on wave direction that decided the ship's movement. By analyzing times series of measured wave elevation using Eqs. (5)–(10), wave spectral density and significant wave height were determined.

Figs. 29 and 30 show wave spectral density in various wave directions for SS2 and SS3 under zero-speed conditions. It could be seen

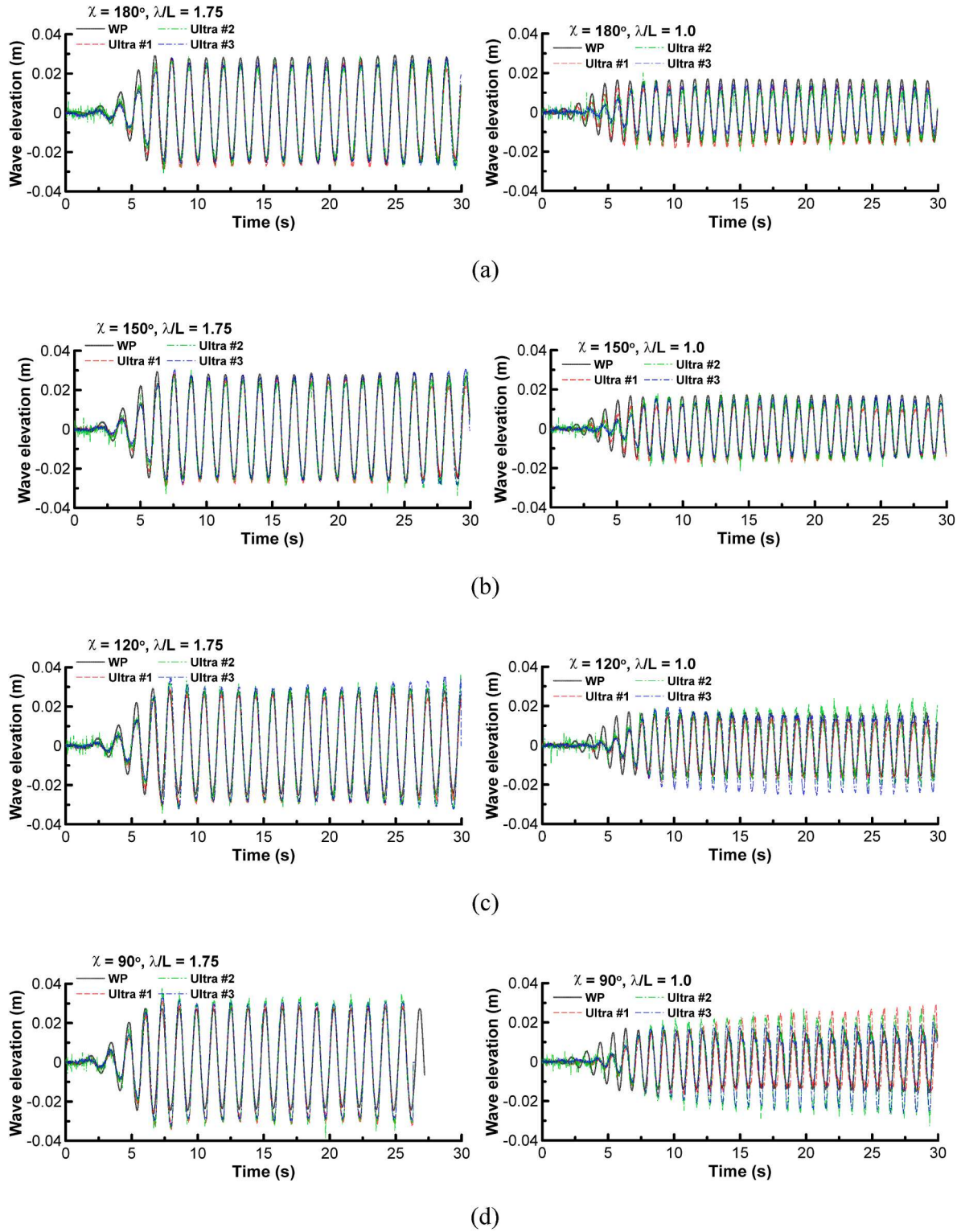


Fig. 21. Measured wave elevation in inner region between demi-hulls at zero speed in regular waves: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$; (e) $\chi = 60^\circ$; (f) $\chi = 30^\circ$; (g) $\chi = 00^\circ$.

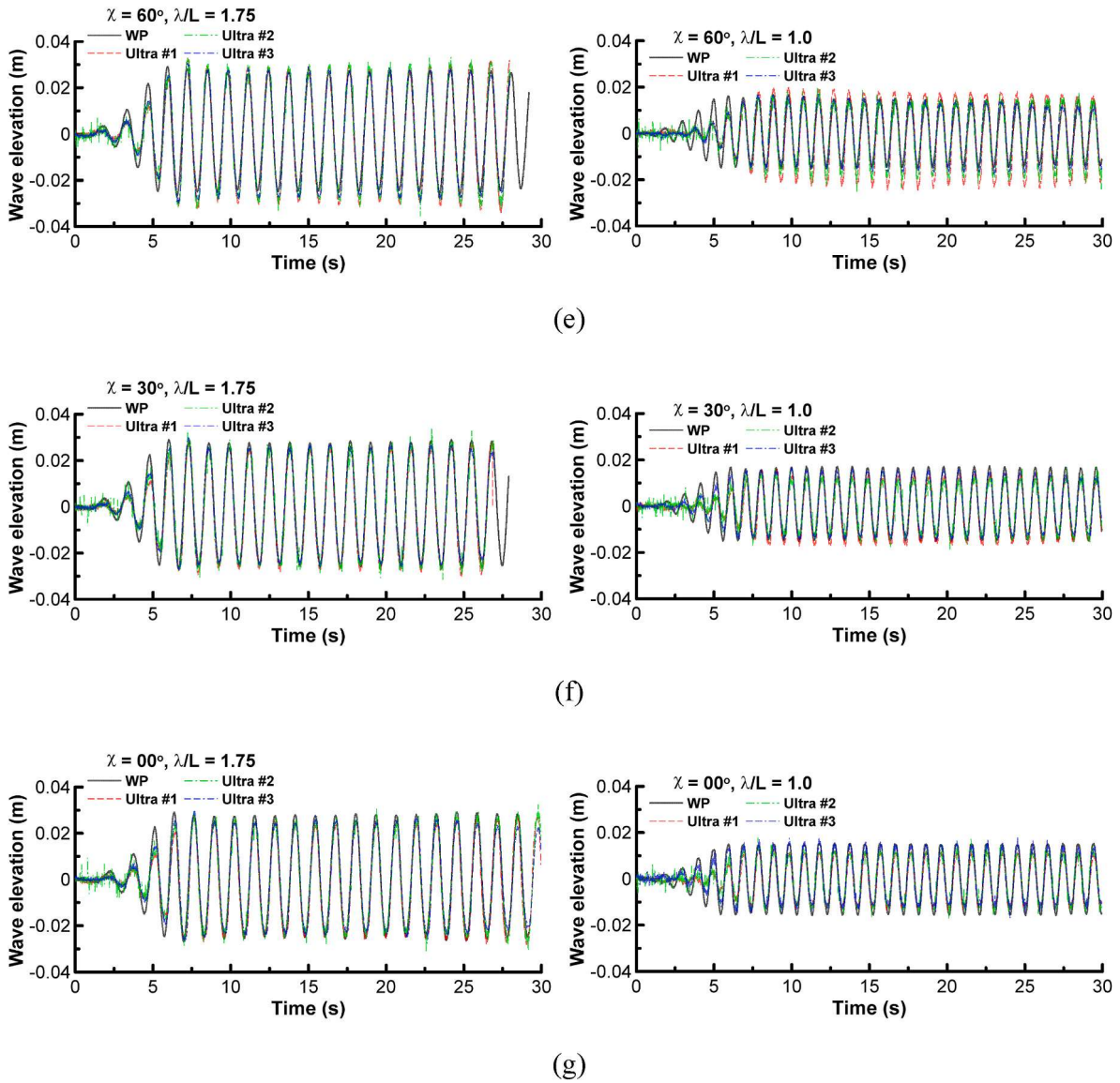


Fig. 21. (continued).

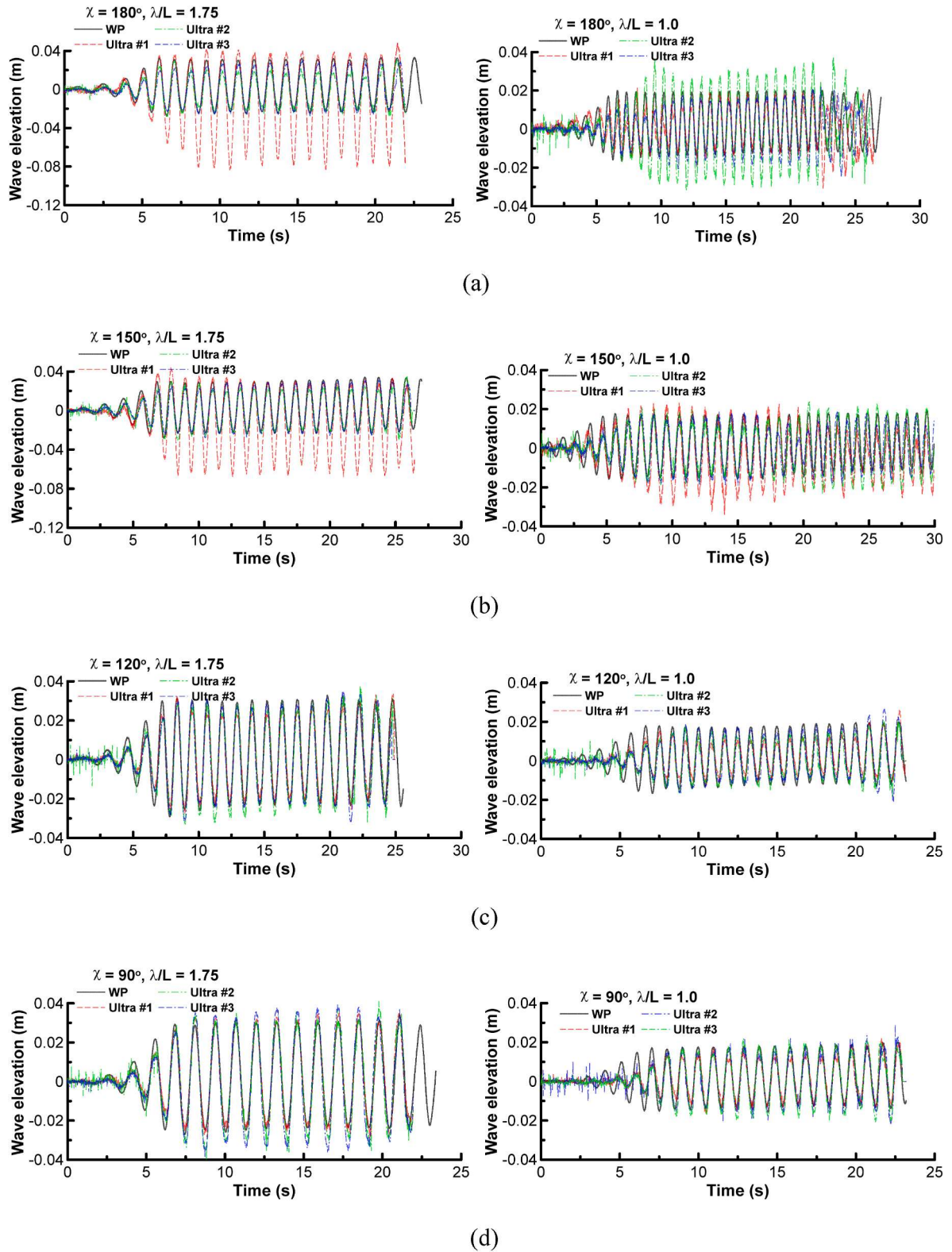


Fig. 22. Measured wave elevation in inner region between demi-hulls at a speed of 0.548 m/s in regular waves: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$; (e) $\chi = 60^\circ$; (f) $\chi = 30^\circ$; (g) $\chi = 00^\circ$.

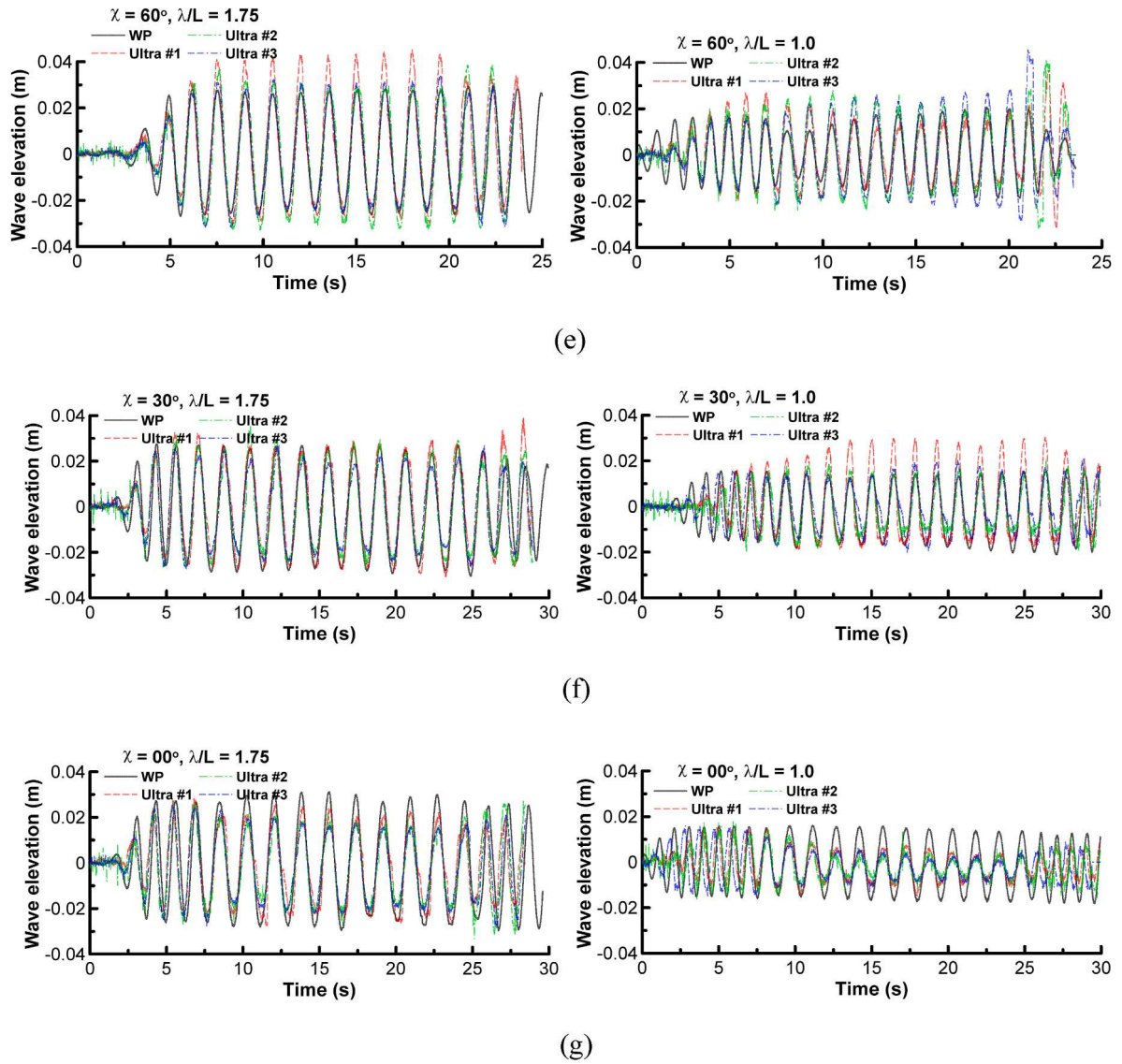


Fig. 22. (continued).

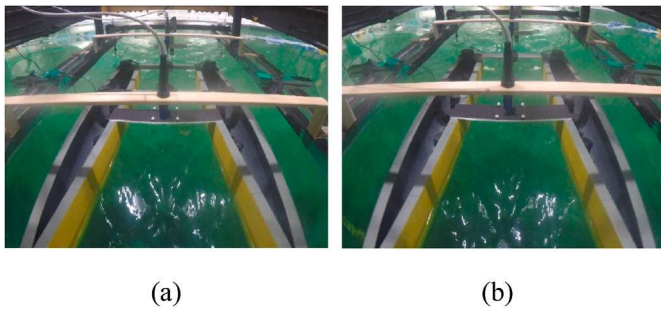


Fig. 23. Capturing wave interference in the inner region between demi-hulls at forward speed and wave direction of 180° during regular waves: (a) $\lambda/L = 2.0$; (b) $\lambda/L = 1.5$.

that magnitudes of spectral densities in the inner region were consistently lower or narrower than those of ocean waves in general for various wave directions of both SS2 and SS3. According to wave directions and position in the inner region for SS2, the spectral density experienced a slight reduction at quartering waves compared to head waves and beam waves. Additionally, the spectral density at the ultrasonic 1 position was greater than that at the ultrasonic 2 position for the bow waves. Meanwhile, the wave at the ultrasonic 3 position was larger than that at the ultrasonic 1 position for the stern waves. Furthermore, it was observed that the spectral density shape and magnitude were symmetrically in some pairs of wave directions such as 150° – 60° and 120° – 30° , in which waves at ultrasonic 1 and 3 positions were exchanged. The spectral density shape and magnitude at ultrasonic 1 and 3 positions exhibited similarities at wave directions of 180° and 90° . There were huge changes in the spectral shape in all wave directions, especially in wave directions of 150° , 30° , and 0° due to mutual impacts of waves between demi-hulls, a consequence of the ship's movement,

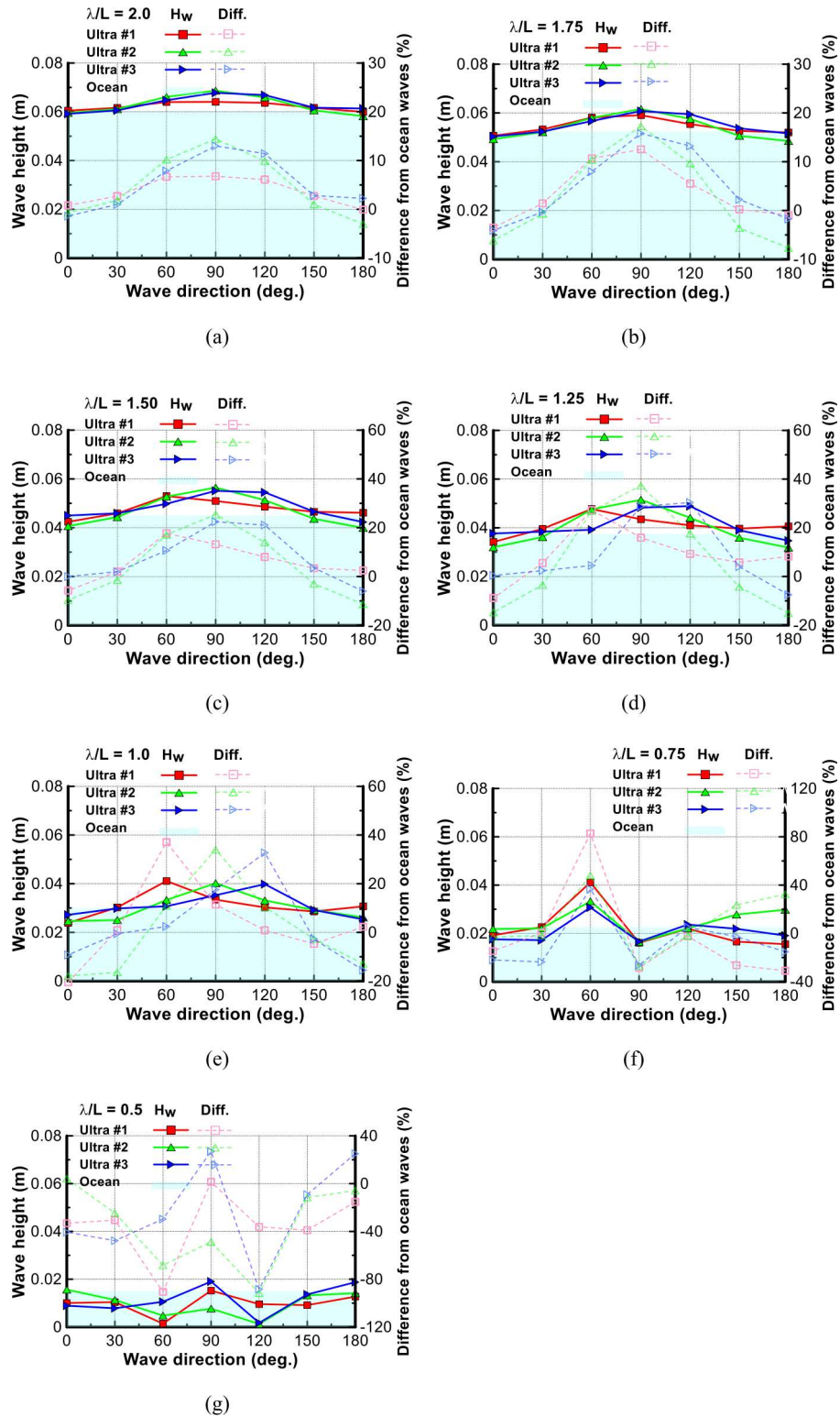


Fig. 24. Wave height at zero speed in regular waves: (a) $\lambda/L = 2.0$; (b) $\lambda/L = 1.75$; (c) $\lambda/L = 1.50$; (d) $\lambda/L = 1.25$; (e) $\lambda/L = 1.0$; (f) $\lambda/L = 0.75$; (g) $\lambda/L = 0.5$.

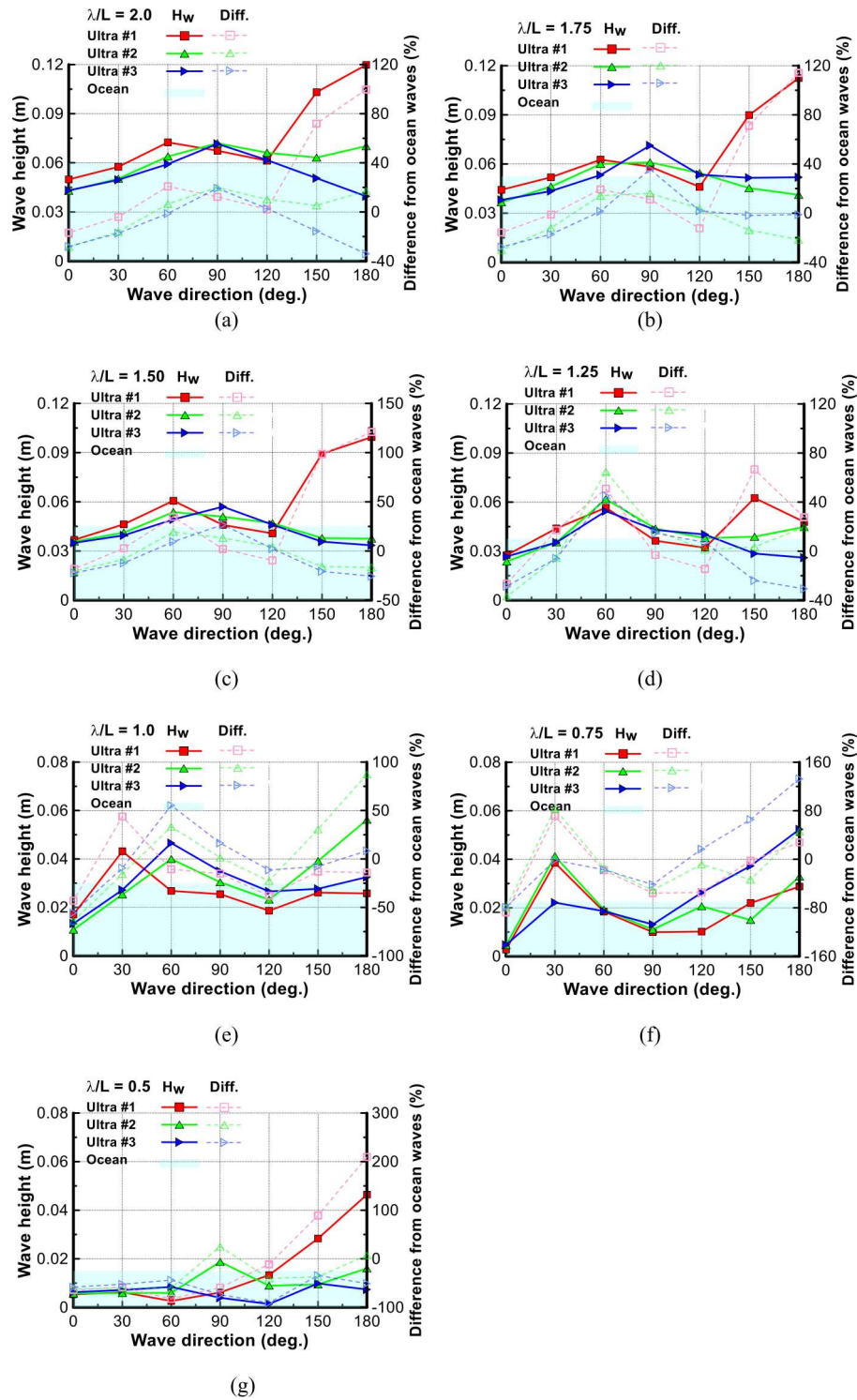
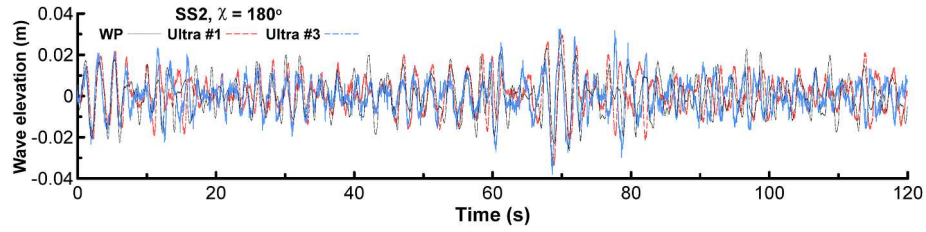
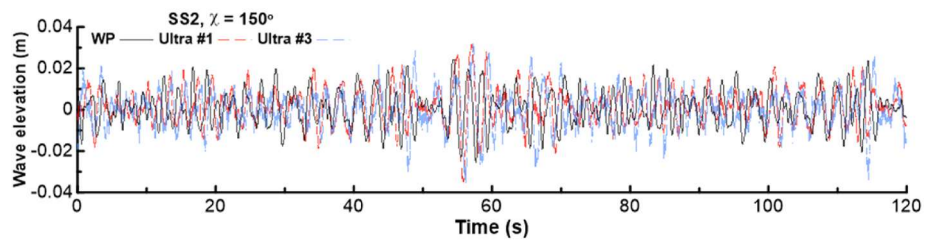


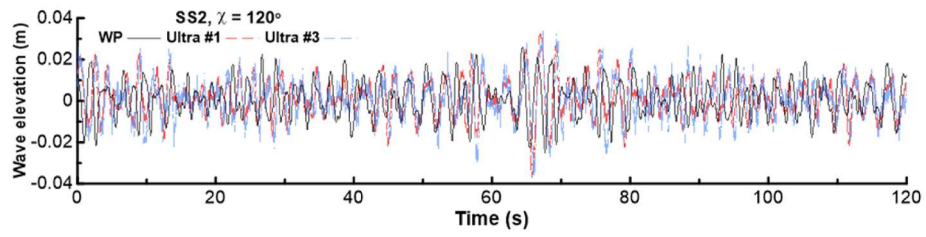
Fig. 25. Wave height at a forward speed of 0.548 m/s in regular waves: (a) $\lambda/L = 2.0$; (b) $\lambda/L = 1.75$; (c) $\lambda/L = 1.50$; (d) $\lambda/L = 1.25$; (e) $\lambda/L = 1.0$; (f) $\lambda/L = 0.75$; (g) $\lambda/L = 0.5$.



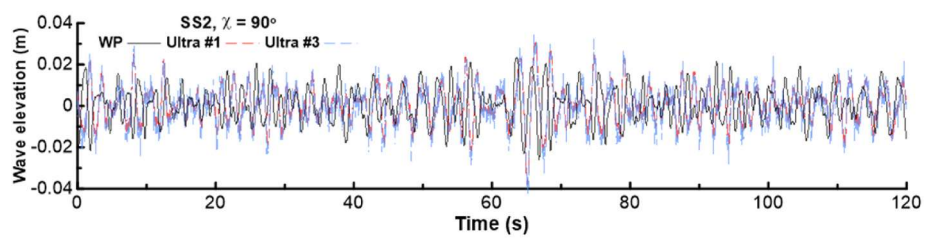
(a)



(b)



(c)



(d)

Fig. 26. Measured wave elevation at SS2 and zero speed: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$.

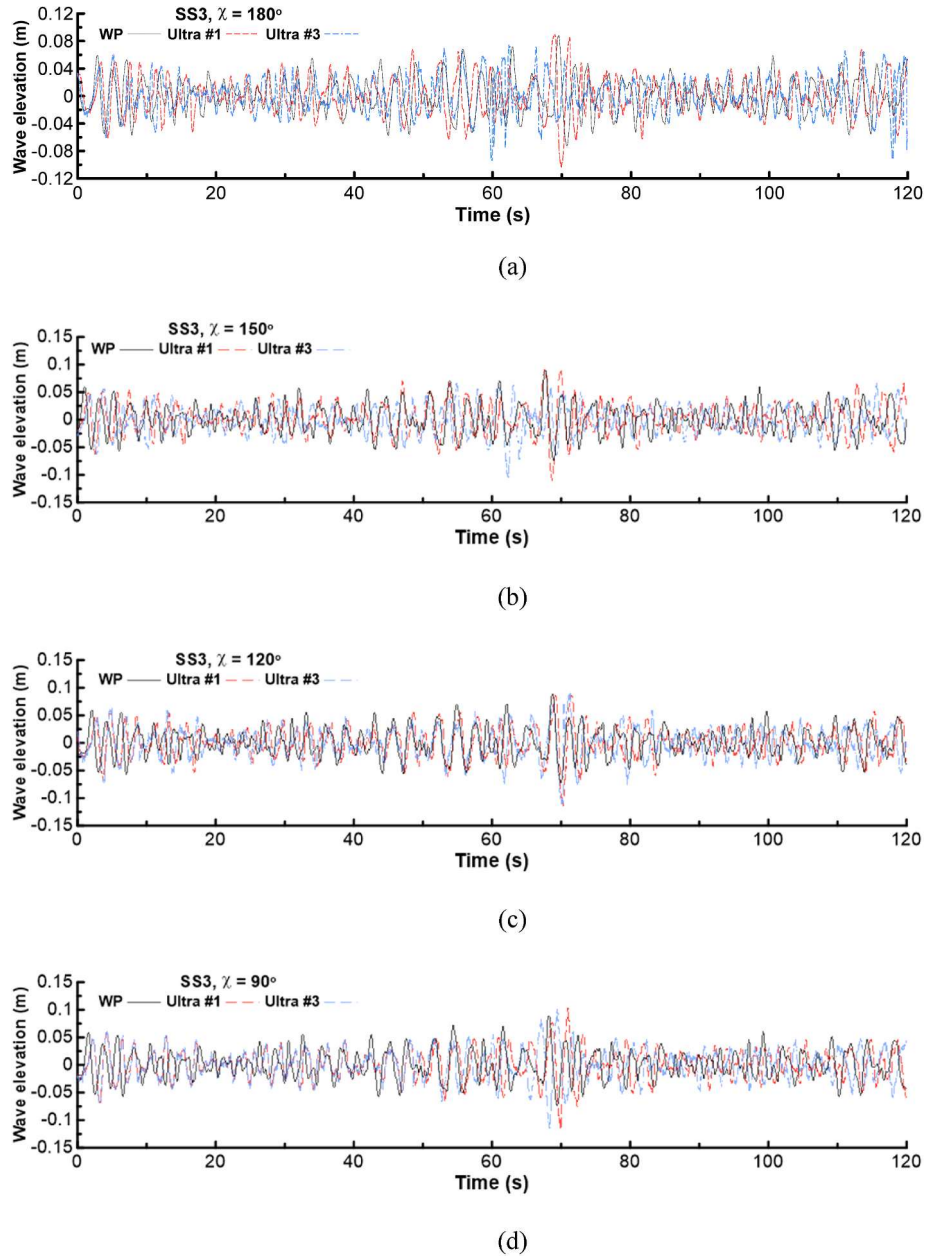


Fig. 27. Measured wave elevation at SS3 and zero speed: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$.

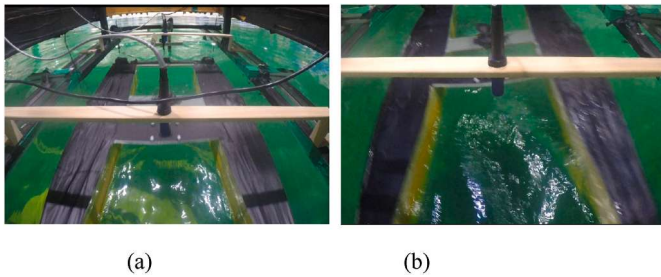


Fig. 28. Capturing a moment of wave interference in the inner region between demi-hulls at zero speed during irregular waves of SS3: (a) $\chi = 180^\circ$; (b) $\chi = 90^\circ$.

resulting in generation of radiation waves from demi-hulls. External waves and radiation waves were then canceled by each other in portion, leading to a reduction of wave height and a change in frequency. These observations became more pronounced at SS3, as the ship's movement was stronger. As shown in Fig. 30, the spectral density shape and magnitude in the inner region changed significantly compared to ocean waves at ultrasonic 1 and 3 positions, especially at wave directions of 180° , 150° , 30° , and 0° . However, it presented similarities at ultrasonic 1 and 3 positions in wave directions of 120° , 90° , and 60° . This was due to mutual impacts of waves between demi-hulls when the ship's movement in sway and yaw was dominant, resulting in waves in the entire the inner region having an equally great interaction.

These disparities in spectral densities demonstrated that the wave energy in the inner region between demi-hulls of catamaran consistently

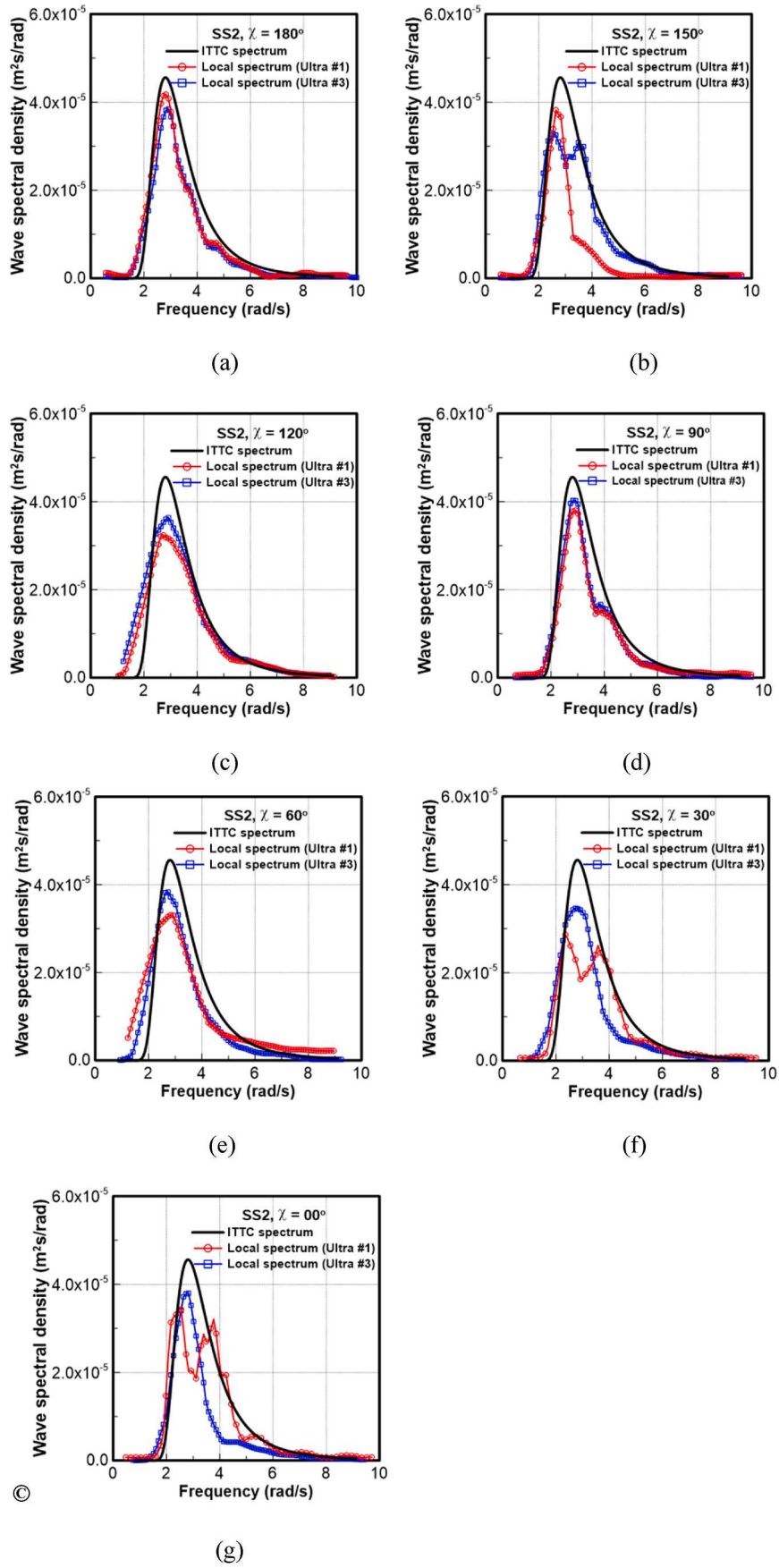


Fig. 29. Wave spectral density at SS2 and zero speed: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$; (e) $\chi = 60^\circ$; (f) $\chi = 30^\circ$; (g) $\chi = 00^\circ$.

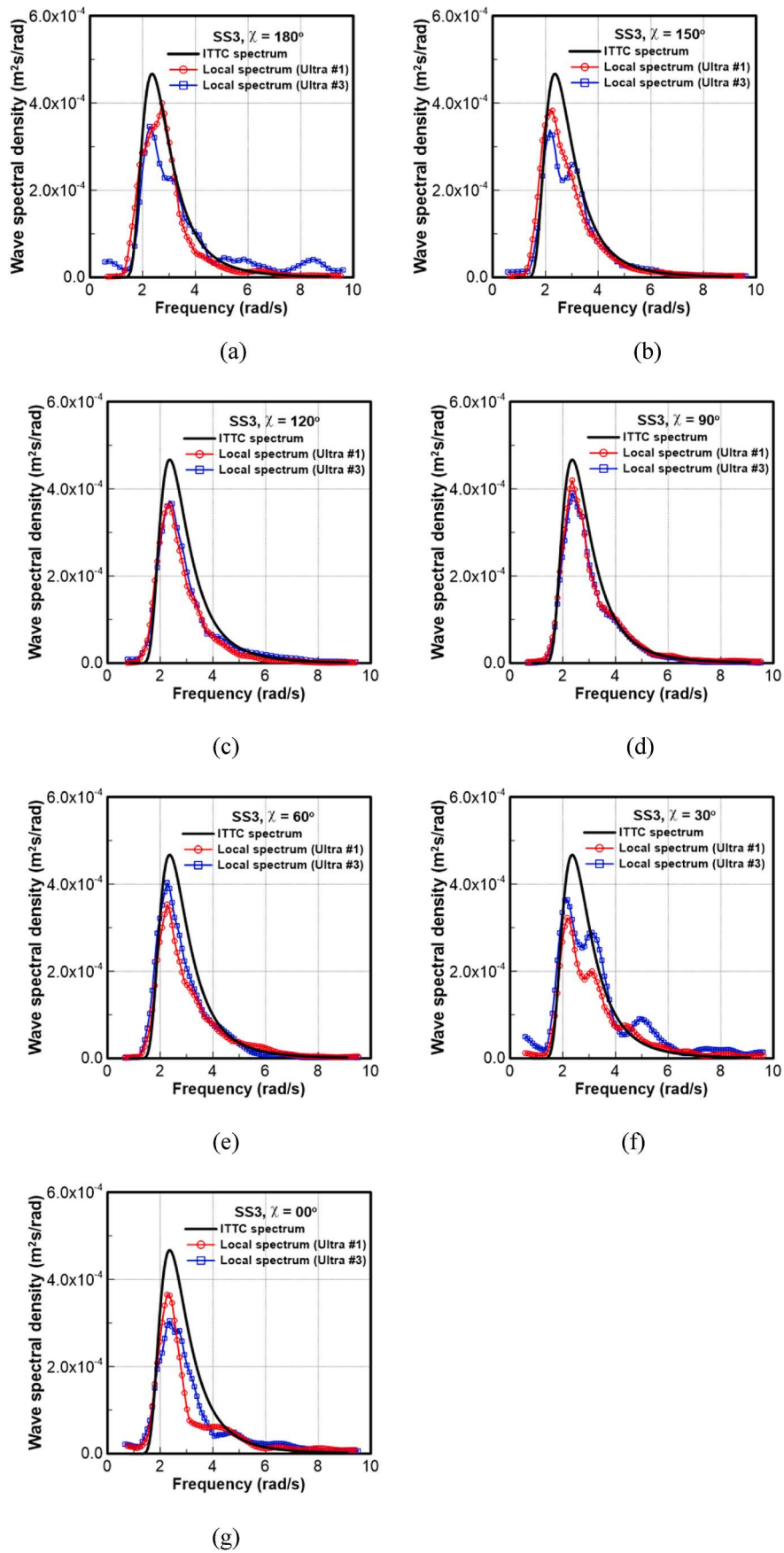


Fig. 30. Wave spectral density at SS3 and zero speed: (a) $\chi = 180^\circ$; (b) $\chi = 150^\circ$; (c) $\chi = 120^\circ$; (d) $\chi = 90^\circ$; (e) $\chi = 60^\circ$; (f) $\chi = 30^\circ$; (g) $\chi = 00^\circ$.

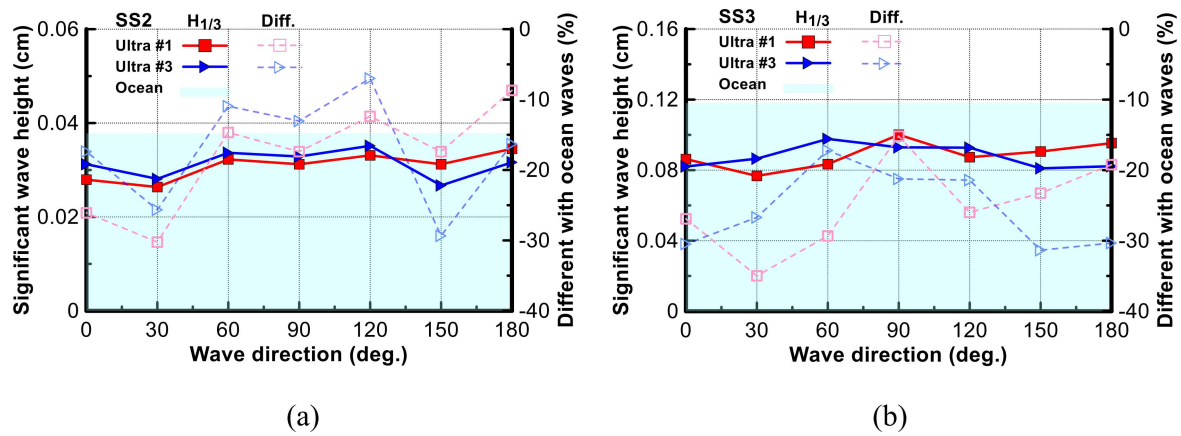


Fig. 31. Significant wave height at zero speed in irregular waves: (a) SS2; (b) SS3.

remained lower than that of ocean waves for irregular waves. Furthermore, spectral density in the inner region was influenced by various factors including sea state, wave directions, and position of the catamaran, which collectively contributed to the variation in wave characteristics in the inner region.

By analyzing wave spectral density, significant wave heights in irregular waves were calculated. Fig. 31 summarizes significant wave height in the inner region at zero speed for various wave directions of the catamaran. Differences from ocean waves were plotted to provide a visualization of the influence of catamaran's presence on significant wave height in the inner region. It could be seen that wave height values in the inner region were significantly lower than those of ocean waves for all sea states and wave directions. As a result, the difference from ocean waves was consistently negative for all cases. When significant wave height of SS2 in Fig. 31a was examined, its value was around 0.03 m for all wave directions, smaller than the significant wave height of ocean waves. The difference with ocean waves of SS2 was observed to be under 20% at wave directions of 180°, 120°, 90°, and 60° for both ultrasonic 1 and 3 positions. Meanwhile, the difference with ocean waves of SS3 exceeded 20% at almost all wave directions as shown in Fig. 31b. Although differences of both SS2 and SS3 were below 35%, those of SS2 were lower than those of SS3. This indicated that differences in significant wave height compared to ocean waves in the inner region were increased at a higher sea state.

Variations of wave spectral density and significant wave height in inner regions between demi-hulls of the catamaran once again confirmed substantial influences of sea state, wave direction, and measurement position on wave characteristics in the inner region during irregular waves.

Investigation of wave characteristics in the inner region between demi-hulls of the catamaran provided valuable insights into variations of wave elevation as well as wave height between demi-hulls of catamaran in regular waves and significant wave height in irregular waves. Understanding variations of wave characteristics in the inner region is crucial for assessing and predicting the motion of an AUV during the LAR process right beneath the catamaran under different wave conditions.

4. Concluding remarks

This study investigated wave characteristics in the inner region between demi-hulls of the catamaran by measuring wave elevation through a model test in the wave tank in regular and irregular waves. The measurement was implemented using the ultrasonic sensors in the inner region and the capacitance wave probe in ocean waves with catamaran moved in various wave directions at various speed conditions.

For regular waves, wave height and wave elevation were considered as key parameters. For irregular waves, wave spectral density and significant wave height were parameters representative of wave characteristics. The experimental method and setting were validated by comparing the motion response of the catamaran measured by experiment with those of numerical calculation by the 3D panel method.

For regular waves, wave characteristics in the inner region of the catamaran were investigated for various wavelengths, wave directions, speeds, and measurement positions in the inner region. At zero speed, wave elevation in the inner region closely resembles or slightly surpasses that of ocean waves, especially at long wavelengths. The wave elevation for shorter wavelengths showed a considerable change depending on the measurement position in the inner region which was increased at near stern positions for bow waves and at near bow positions for stern waves. For a forward speed of 0.548 m/s, the wave elevation in the inner region had a substantial variation for both long and shorter wavelengths. Specifically, wave elevation at the near bow position took a deeply concave down at bow waves. The middle body position was smaller than ocean waves for head waves and following waves for long wavelengths. However, wave elevation at the inner region was the greatest at the wave direction of head waves and stern quartering waves for shorter wavelengths. Conversely, wave height was observed to be lowest for following waves. Furthermore, wave height values showed considerable disparities among positions in the inner region and wave directions. This highlighted effects of wavelength, wave direction, speed, and measurement positions in the inner region on wave characteristics under regular waves.

Wave characteristics in the inner region of the catamaran under irregular waves were examined for SS2 and SS3 conditions at various wave directions. Wave elevation in the inner region was measured within 10 min at zero speed. Wave elevation was then analyzed using FFT to obtain wave spectral density and significant wave height. The spectral density and significant wave height in the inner region were observed to be consistently lower than those of ocean waves. This was due to a large movement of the ship, resulting in generation of radiation waves from demi-hulls. These waves and external waves were canceled by each other in portion, leading to a reduction in wave height and a change in frequency in the inner region. This change depended on wave direction and positions in the inner region. Additionally, differences of significant wave height compared to ocean waves in the inner region were increased at a higher sea state.

Although wave elevation was measured at certain positions relative to the carriage due to limitations in the size of the model ship, it still fell in the inner region between demi-hulls, which provided valuable insights into variations of wave elevation as well as wave height between demi-hulls of the catamaran for regular waves and significant wave

height for irregular waves. Understanding these variations is crucial for accurately assessing and predicting the motion of an AUV during the LAR process under various wave conditions. By considering the wave characteristics in the inner region, the RMS (Root Mean Square) motion response of AUV beneath USV and relative motion between USV and AUV will be calculated in further studies, which informed decisions can be made to enhance the efficiency and safety of the AUV throughout the LAR process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by Unmanned Vehicles Core Technology Research and Development Program through the National Research Foundation of Korea (NRF) and Unmanned Vehicle Advanced Research Center (UVARC) funded by Ministry of Science and ICT, the Republic of Korea (2020M3C1C1A02086323), and by the National Research Foundation of Korea(NRF) grant funded by Korea government (MSIT) (NRF-2022R1A2C1093055).

References

- Allen, B., Austin, T., Forrester, N., Goldsborough, R., Kukulya, A., Packard, G., Purcell, M., Stokey, R., 2006. Autonomous docking demonstrations with enhanced REMUS technology. *Oceans* 2006. <https://doi.org/10.1109/OCEANS.2006.306952>. September 2014.
- Aggarwal, A., Pákozdi, C., Bihs, H., Myrhaug, D., Chella, M.A., 2018. Free surface reconstruction for phase accurate irregular wave generation. *J. Mar. Sci. Eng.* 6, 105. <https://doi.org/10.3390/jmse6030105>.
- Bishop, C.T., Donelan, M.A., 1987. Measuring waves with pressure transducers. *Coast. Eng.* 11, 309–328. [https://doi.org/10.1016/0378-3839\(87\)900317](https://doi.org/10.1016/0378-3839(87)900317).
- Bouscasse, B., Broglio, R., Stern, F., 2013. Experimental investigation of a fast catamaran in head waves. *Ocean. Eng.* 72, 318–330. <https://doi.org/10.1016/j.oceaneng.2013.07.012>, 2013.
- Broglio, R., Zaghi, S., Di Mascio, A., 2011. Numerical simulation of interference effects for a high-speed catamaran. *J. Mar. Sci. Technol.* 16 (3), 254–269. <https://doi.org/10.1007/s00773-011-0132-3>.
- Broglio, R., Jacob, B., Zaghi, Stefano, Stern, F., Olivieri, A., 2014. Experimental investigation of interference effects for high-speed catamarans. *Ocean. Eng.* 76, 75–85. <https://doi.org/10.1016/j.oceaneng.2013.12.003>, 2014.
- Cavaleri, C., 1980. Wave measurement using pressure transducer. *Oceanol. Acta* 3, 339–346.
- Castiglione, T., Stern, F., Bova, S., Kandasamy, M., 2011. Numerical investigation of the seakeeping behavior of a catamaran advancing in regular head waves. *Ocean. Eng.* 38, 1806–1822. <https://doi.org/10.1016/j.oceaneng.2011.09.003>, 2011.
- Gao, W., Liu, W., Li, F., Hu, Y., 2021. Analysis and validation of ultrasonic probes in liquid level monitoring systems. *Sensors* 21, 1–13. <https://doi.org/10.3390/s21041320>, 2021.
- ITTC, 2021. *Seakeeping Experiments. Recommended Procedures and Guidelines*, vol. 7, 5-02-07-02.1.
- Jeon, M., Mai, T.L., Yoon, H.K., Kim, D., 2021. Estimation of wave-induced steady force using system identification, model tests, and numerical approach. *Ocean. Eng.* 233, 109207. <https://doi.org/10.1016/j.oceaneng.2021.109207>.
- Jung, H.C., Koo, W., 2021. An experimental study on wave absorber performance of combined punching plate in a two-dimensional mini wave tank. *J. Ocean Eng. Technol.* 35, 113–120. <https://doi.org/10.26748/KSOE.2021.006>.
- Kim, D.J., Yun, K., Park, J.Y., Yeo, D.J., Kim, Y.G., 2019. Experimental investigation on turning characteristics of KVLCC2 tanker in regular waves. *Ocean. Eng.* 175 (February), 197–206. <https://doi.org/10.1016/j.oceaneng.2019.02.011>.
- Kim, D.J., Choi, H., Yun, K., Yeo, D.J., Kim, Y.G., 2022. Experimental study on turning characteristics of KVLCC2 tanker in long-crested irregular waves. *Ocean. Eng.* 244 (June 2021), 110362 <https://doi.org/10.1016/j.oceaneng.2021.110362>.
- Kim, H., Jeon, C., Kim, K., Seo, J., 2023. Uncertainty assessment of wave elevation field measurement using a depth camera. *J. Mar. Sci. Eng.* 11, 657. <https://doi.org/10.3390/jmse11030657>.
- Lewis, E.V., 1989. *Principles of Naval Architecture*. The Society of Naval Architectures and Marine Engineers.
- Liu, Z., Zhang, X., Meng, Y., Wang, L., 2021. Numerical calculation of the resistance of catamarans at different distances between two hulls. In: 3rd International Conference on Civil, Architecture and Urban Engineering, vol. 283.
- Mai, T.L., Vo, A.K., Cho, A., Yoon, H.K., 2023. Experimental study of wave induced motions and loads on catamaran in regular and irregular waves. In: The 33rd International Ocean and Polar Engineering Conference. Ottawa, Canada.
- Mandru, A., Pacuraru, F., 2020. Experimental study on catamaran hydrodynamics. *IOP Conf. Ser. Mater. Sci. Eng.* 916, 2020.
- Milanov, E., Chotukova, V., Stern, F., 2012. System Based Simulation of Delft372 Catamaran Maneuvering Characteristics as Function of Water Depth and Approach Speed. In: 29th Symposium on Naval Hydrodynamics Gothenburg. Sweden, pp. 26–31.
- Milanov, E., Zlatev, Z., Chotukova, V., Stern, F., 2011. Analysis of inherent course stability of a high-speed catamaran in deep and shallow water. *International Shipbuilding Progress* 58, 83–96.
- Nam, H.S., Park, D.M., Cho, S.K., Hong, S.Y., 2022. Analysis of relative wave elevation around semi-submersible platform through model test: focusing on comparison of wave probe characteristics. *J. Ocean Eng. Technol.* 36, 1–10. <https://doi.org/10.26748/KSOE.2021.094>.
- Oh, S., Jung, J.H., Cho, S.K., 2020. Higher-order spectral method for regular and irregular wave simulations. *J. Ocean Eng. Technol.* 34, 406–418. <https://doi.org/10.26748/KSOE.2020.052>.
- Pai, S., Guerrini, P., Potter, J., Maguer, A., Chitre, M., Bartolomeo, V.S., 2007. *Autonomous Initial Capture System for AUV Recovery*. December 2013.
- Pai, S., Guerrini, P., Potter, J., Maguer, A., Chitre, M., Bartolomeo, V.S. 2012. *Autonomous Initial Capture System for AUV Recovery*.
- Pandey, J., Hasegawa, K., 2016. Study on turning manoeuvre of catamaran surface vessel with a combined experimental and simulation method. *IFAC-PapersOnLine* 49 (23), 446–451. <https://doi.org/10.1016/j.ifacol.2016.10.446>.
- Sarda, E., Dhanak, M., 2013. Unmanned recovery of an AUV from a surface platform manuscript, oceans '13 MTS/IEEE San Diego. *OCEANS 2013 MTS/IEEE - San Diego: An Ocean Common* 1–6. <https://doi.org/10.23919/OCEANS.2013.6741386>.
- Sarda, E.I., Dhanak, M.R., 2017. A USV-based automated launch and recovery system for AUVs. *IEEE J. Ocean. Eng.* 42 (1), 37–55. <https://doi.org/10.1109/JOE.2016.2554679>.
- Sarda, E.I., Dhanak, M.R., 2019. Launch and recovery of an autonomous underwater vehicle from a station-keeping unmanned surface vehicle. *IEEE J. Ocean. Eng.* 44 (2), 290–299. <https://doi.org/10.1109/JOE.2018.2867988>.